

High k gate dielectrics and metal gates

Motivation

Material properties

Process Integration

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Course content

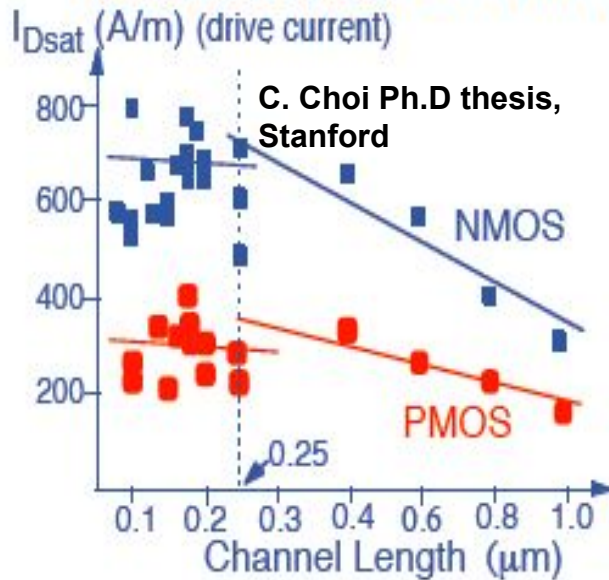
2

- Review of CMOS scaling. Problems with traditional (Dennard) scaling.
- Basic Quantum mechanics and Silicon band structure
- Mobility enhancement techniques (introduced in high volume manufacturing in 2003)
- High k gate dielectrics and metal gates (introduced in 2007)
- FinFETs (introduced in 2012)
- High mobility channel materials (introduced in 2011)
- Nanosheet transistors (introduced in 2024 Samsung Galaxy Watch7)
- Layout dependent effects (became a serious issue since late 2000s)

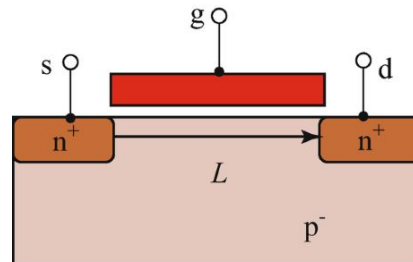
Three horsemen of the apocalypse

3

Constant OFF current Limit



Saturation of I_{dsat}
@ constant I_{off}



IEEE ELECTRON DEVICE LETTERS, VOL. 18, NO. 5, MAY 1997

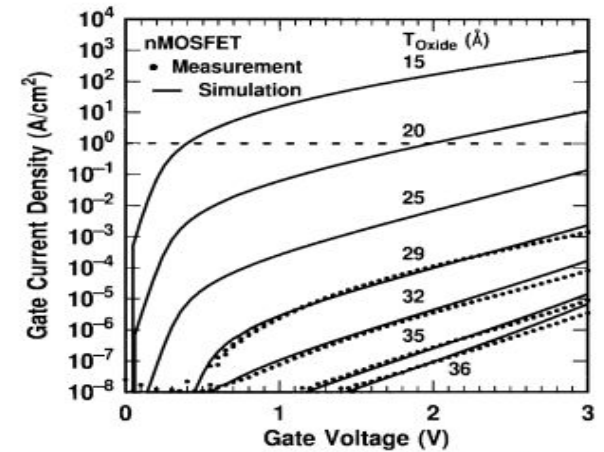
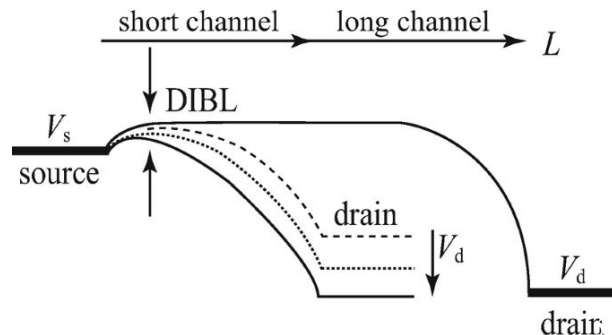


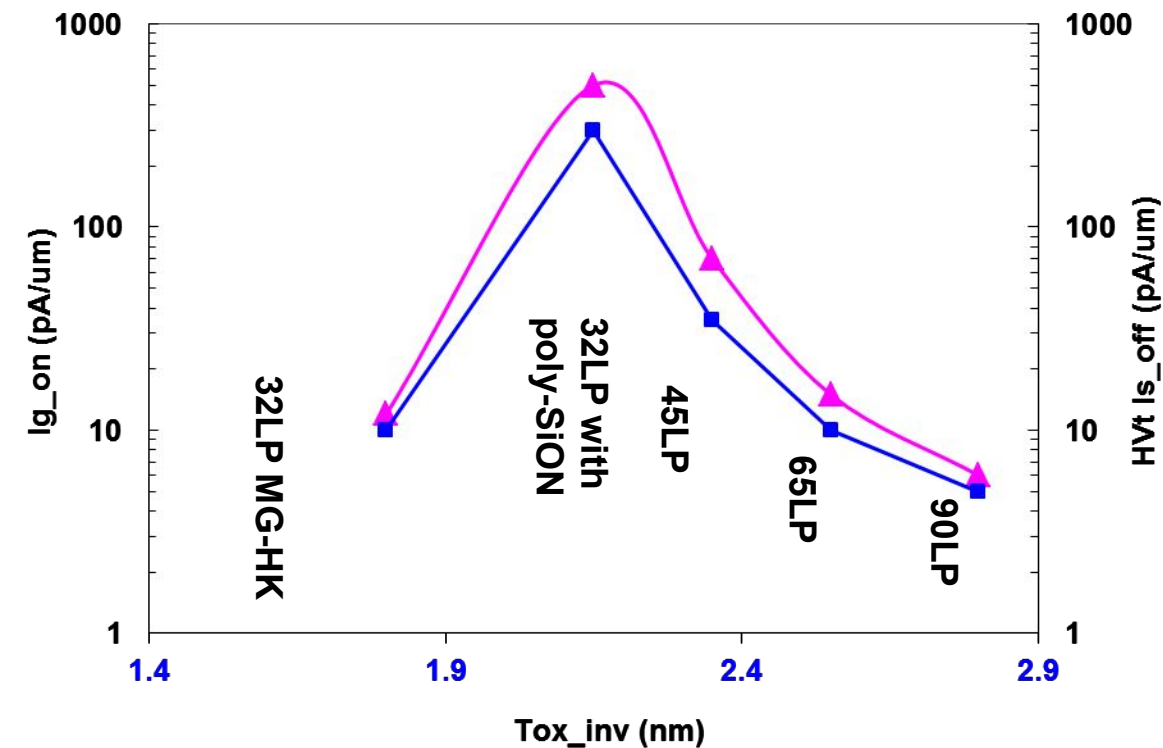
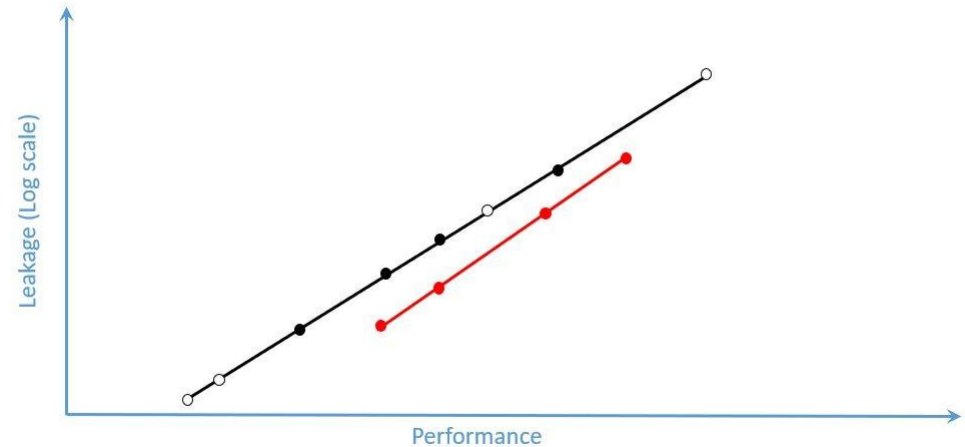
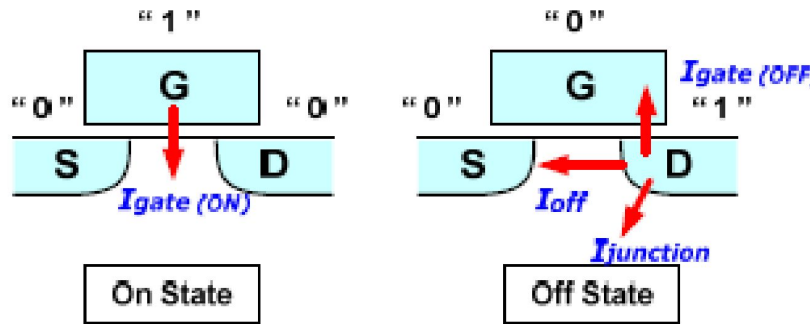
Fig. 2. Measured and simulated I_G-V_G characteristics under inversion conditions of four nMOSFET's. The dotted line indicates the 1 A/cm² limit for leakage current as discussed in the text.

Gate Leakage current
due to tunneling



Worsened gate control

Low Power Leakage Challenge

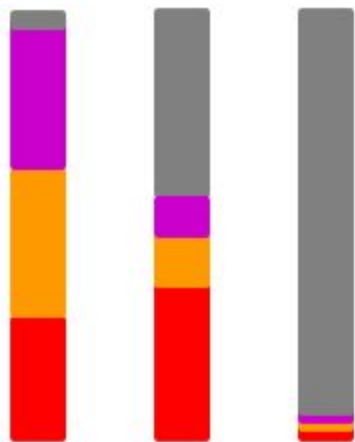


- Gate leakage will limit the offer of HVT low leakage device.
- HK-MG: Enables significant Tox_inv scaling but maintains low gate leakage

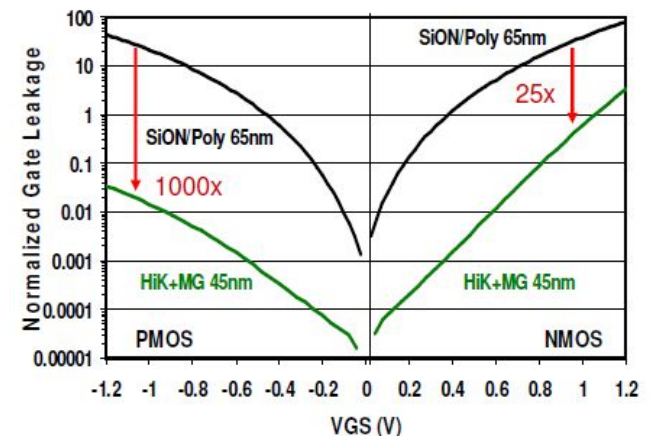
Gate Leakage Contribution to Power Varies Widely by Application

- High-Speed (Server/Games) $I_{gate} \sim 2\%$ of total Leakage Power Negligible benefit.
- General-Purpose (ASIC/Foundry non-LP) $\sim 5\text{-}10\%$ of total Leakage Power – Modest Benefit
- Low-Power (mobile, apps. w/ deep-sleep mode, low-temperature use): $\sim 50\%$ of total Leakage Power-Significant Benefit

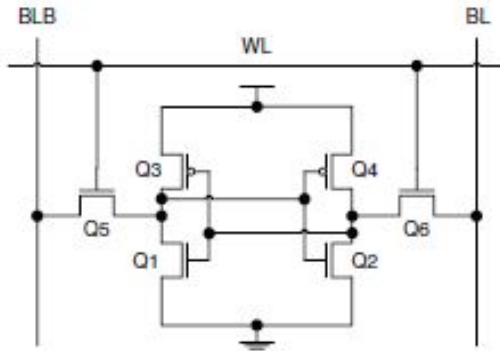
130nm 100nm 65nm



Assenmacher 2003

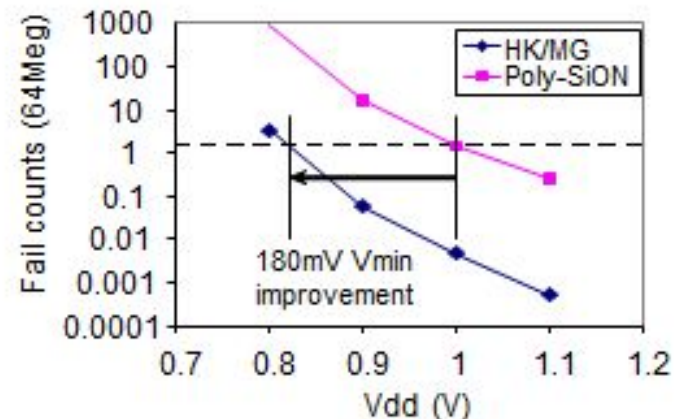
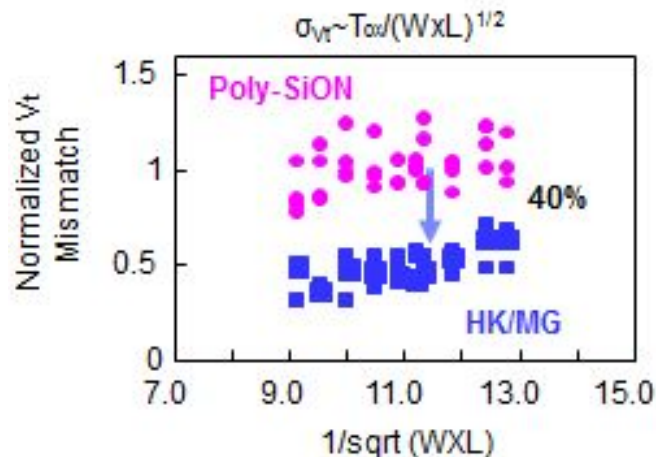


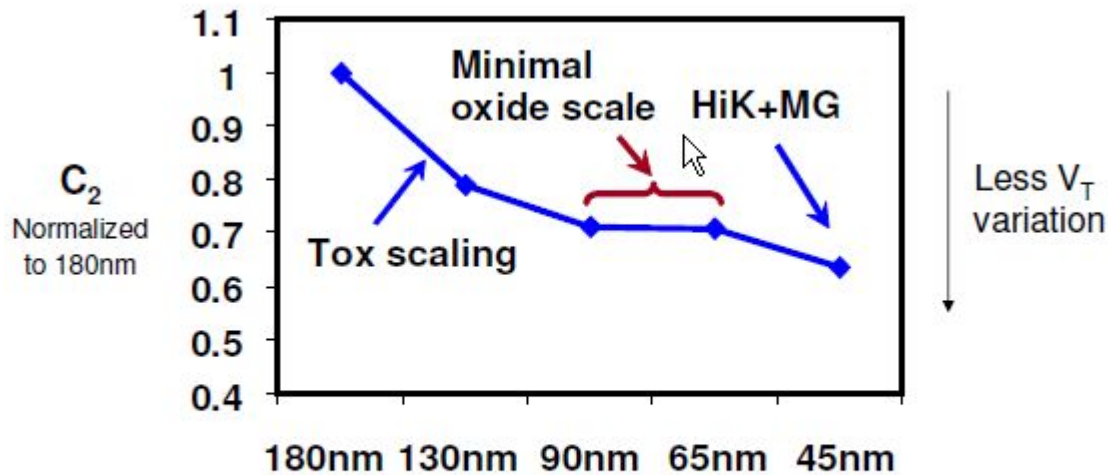
SRAM Area Scaling



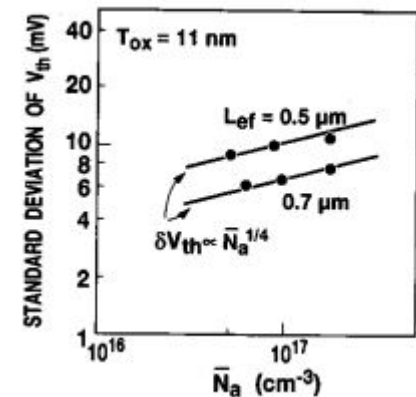
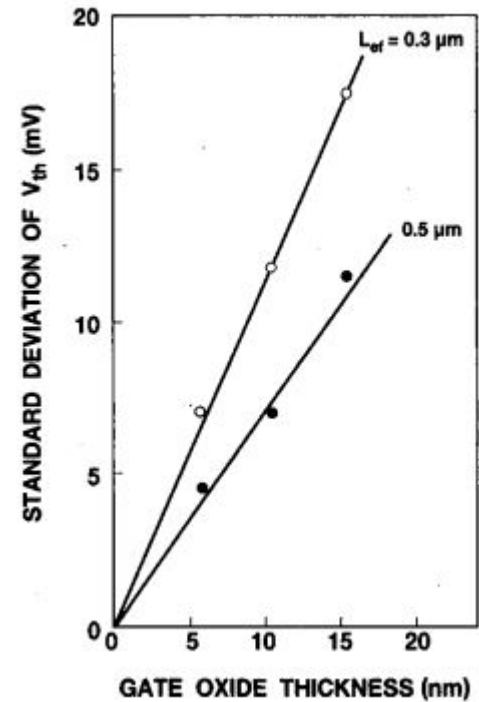
- Low V_{dd} yield is determined by L-R mismatch of SRAM bit cell
- Random dopant fluctuations decide amount of V_T mismatch
- Tox scaling can improve V_T mismatch

$$\sigma V_{T_{\text{tran}}} = \left(\frac{\sqrt[4]{4q^3 \epsilon_{si} \phi_B}}{2} \right) \cdot \frac{T_{ox}}{\epsilon_{ox}} \cdot \left(\frac{\sqrt[4]{N}}{\sqrt{W_{eff} \cdot L_{eff}}} \right) =$$





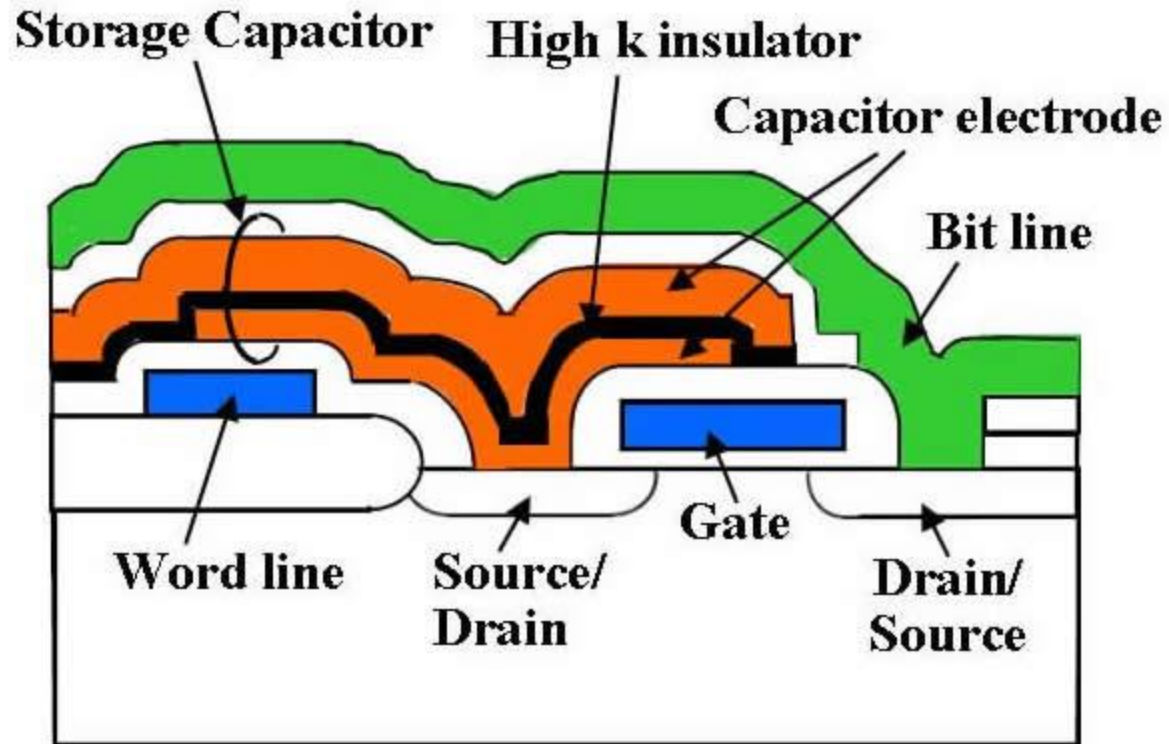
$$\sigma V_{T_{ran}} = \left(\frac{\sqrt[4]{4q^3 \epsilon_{si} \phi_B}}{2} \right) \left(\frac{T_{ox}}{\epsilon_{ox}} \right) \left(\frac{\sqrt[4]{N}}{\sqrt{L_{eff} \cdot Z_{eff}}} \right) = \frac{1}{\sqrt{2}} \left(\frac{C_2}{\sqrt{L_{eff} \cdot Z_{eff}}} \right)$$



T.Mizuno, IEEE Trans on ED, Nov 1994

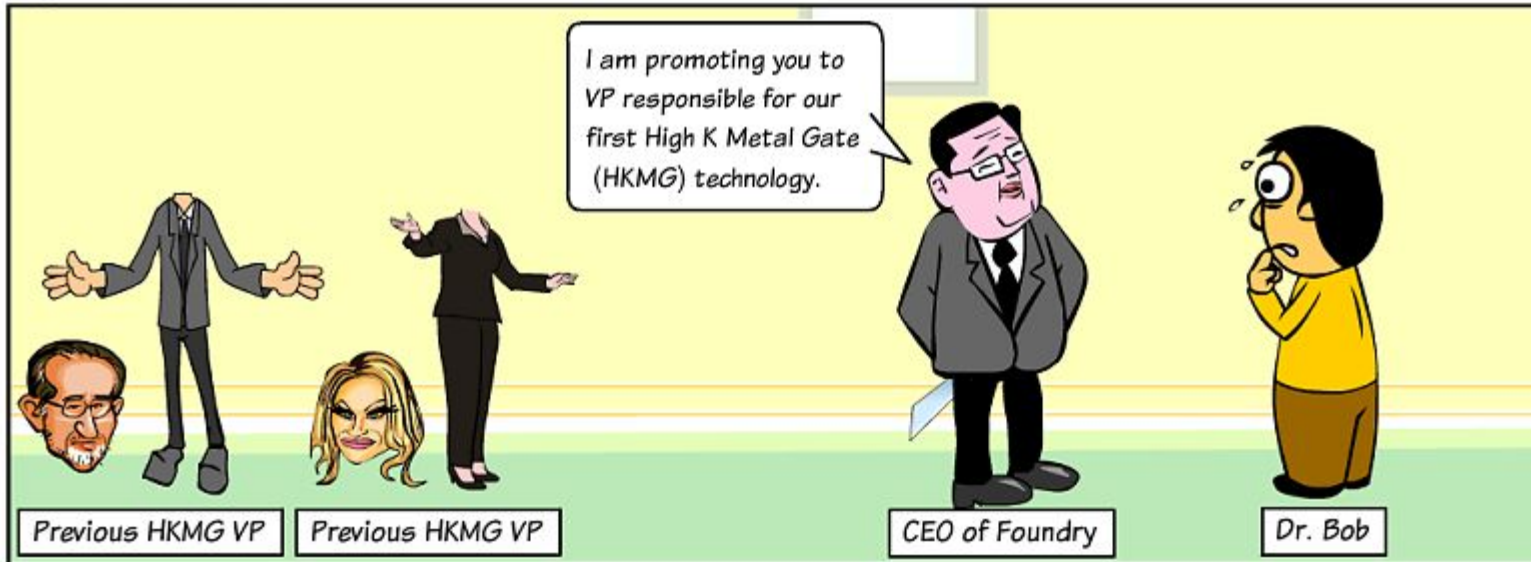
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High k dielectrics in CMOS (DRAM)



Life on a hot seat

HKMG: A HOT POTATO

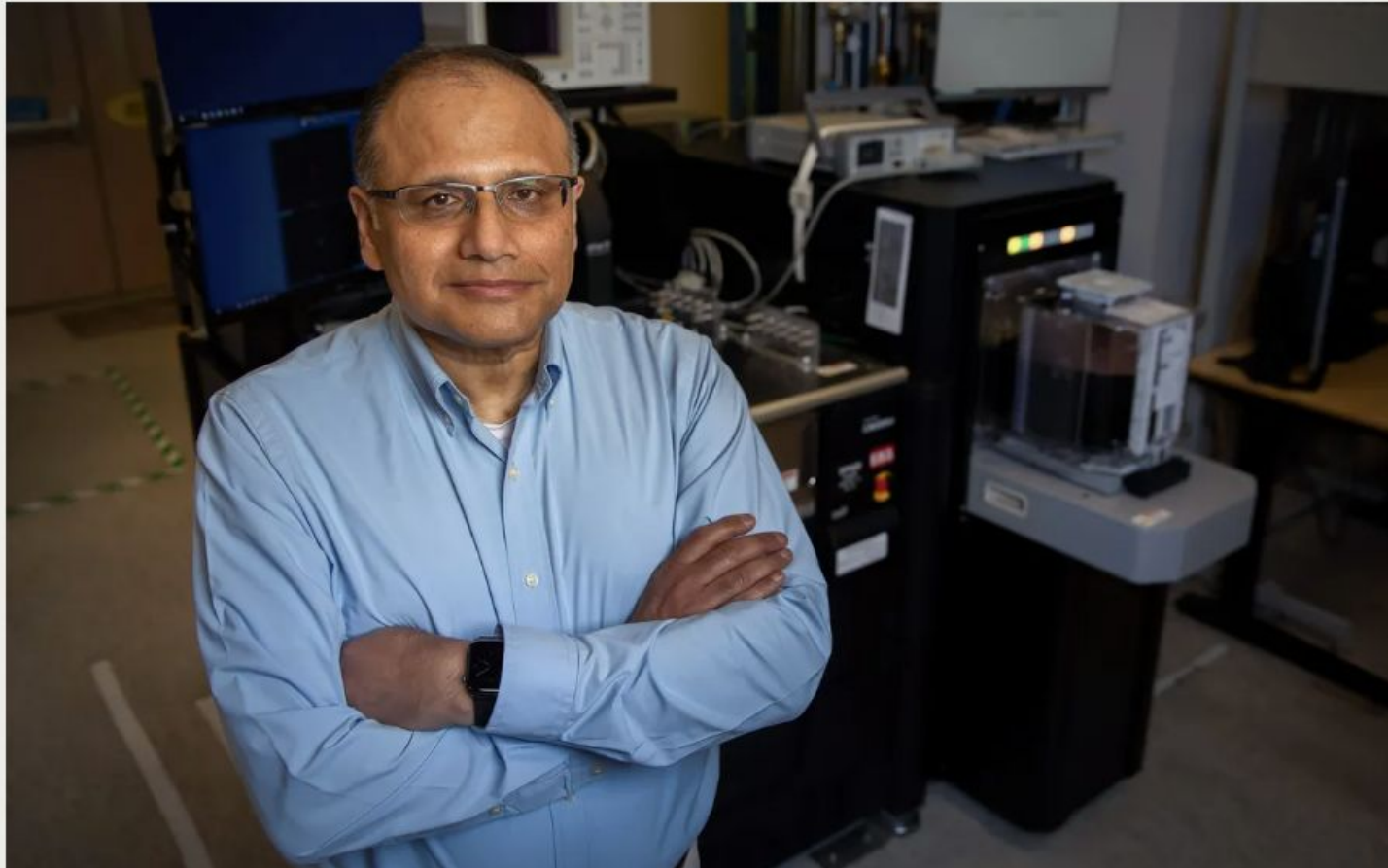


“Mr. Transistor’s” Most Challenging Moment > Intel’s Tahir Ghani says nothing was riskier than the 45-nanometer node tech

BY SAMUEL K. MOORE | 29 JAN 2025 | 5 MIN READ | 

Samuel K. Moore is IEEE Spectrum’s semiconductor editor.

<https://spectrum.ieee.org/transistor-history-2670915657>

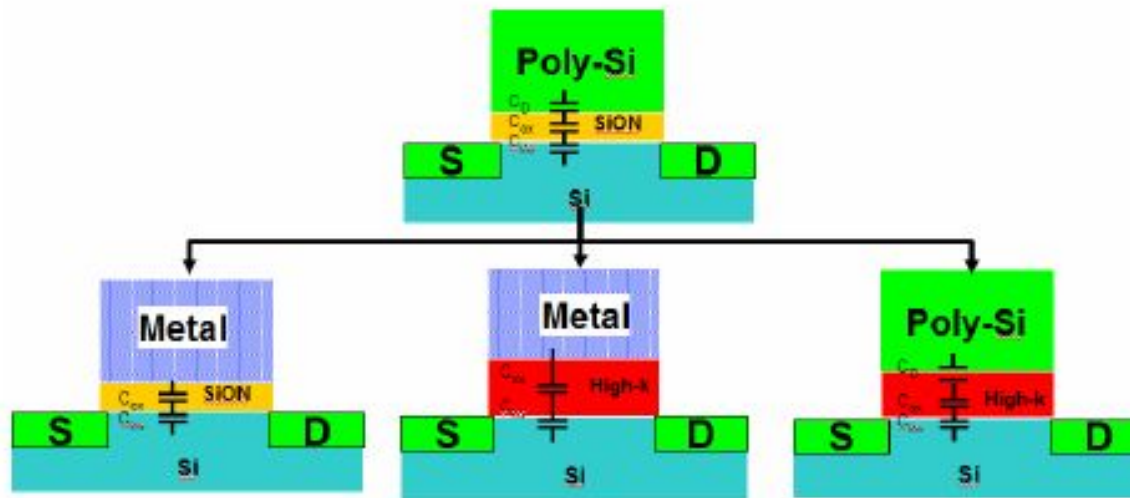


Tahir Ghani—known to his colleagues at Intel as “Mr. Transistor”—says that implementing the tech for 45-nanometer nodes was the riskiest moment of his career. INTEL CORP.

High-k / metal gate: requirements and short history

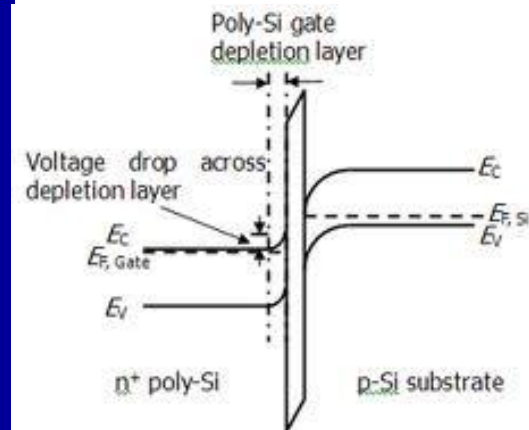
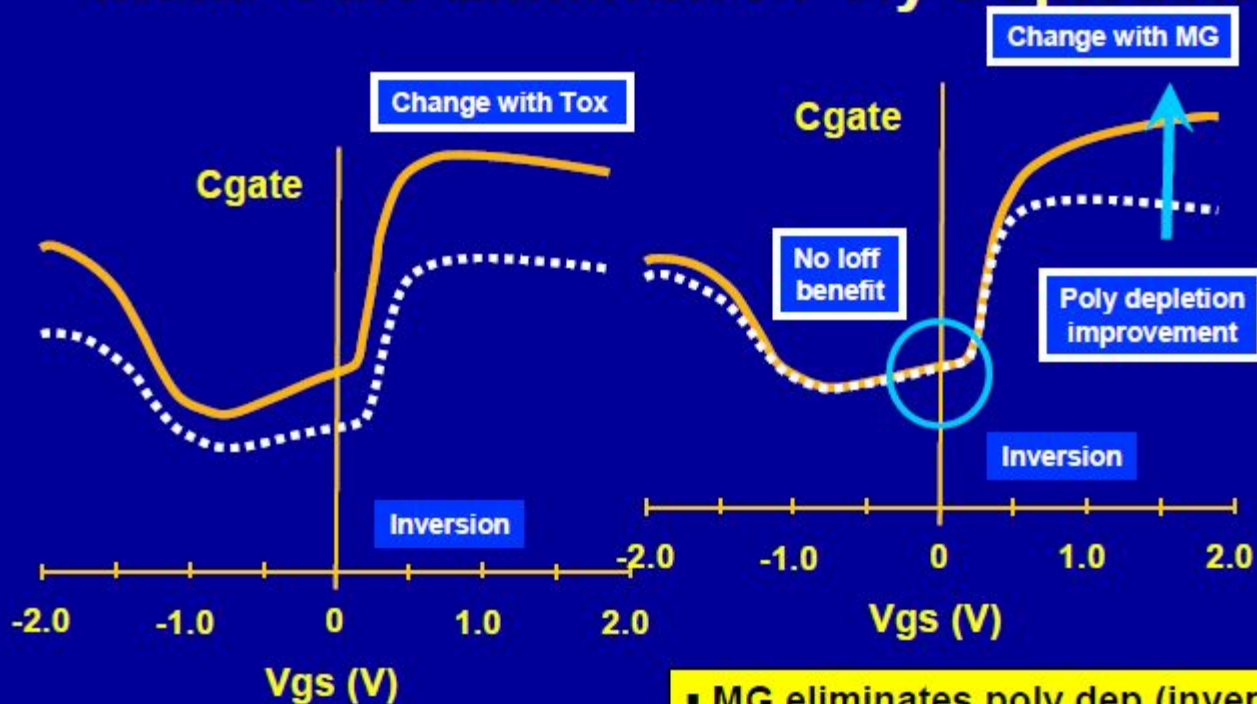
- Requirements of gate stack with high-k gate dielectric
 - Must enable $<1.0\text{nm}$ Equivalent Oxide Thickness (EOT).
 - Must reduce gate leakage.
 - Must have at least equivalent reliability as SiO_2 .
 - Want mobility to be $\sim 90\%$ of SiO_2
 - Must enable gate length scaling
 - Manufacturable in high volume
- Abbreviated history of high-k
 - First plan was to introduce high-k with polysilicon gate electrodes, but most high-k materials for a defect when bonded with silicon $\square$ fermi level pinning
 - Second plan: move to metal gate electrodes with high-k to avoid
- Additional requirements with high-k + metal gate
 - Must identify 2 metal gate materials: 1 metal with workfunction of 4.1eV (n+ poly) and 1 metal with workfunction of 5.2eV (p+ poly).
 - Or must identify 1 metal where workfunction can be modulated between 4.1eV to 5.1eV .

Gate stack replacement



- 1: SiO₂ → High-k: a) "High-k" means anything higher than SiON, $k \sim 5$, $C = \epsilon_0 \kappa A / t$,
 b) Improved EOT scaling (increased cap.) for short channel control
 Capacitance is linearly dependent on thickness
 c) Gate leakage reduction-exponentially dependent on thickness
- 2: Poly-Si → Metal : a) Metals eliminate poly-Si depletion: Increased Cap.
 ($\Delta T_{inv} = -0.4 \text{ nm}$)

Metal Gate Eliminates Poly Depletion



True " T_{ox} " change
Capacitance benefit everywhere
Both I_{dsat} and SCE improve

- MG eliminates poly dep (inversion)
 - Increases gate E -field
 - Larger Q_{inv} → High I_{dsat}
- T_{ox} (inv) scales significantly
- BUT! $T_{ox}(e)$ does not impact SCE

Periodic table

IA																	O			
*	H	IIA												III A	IV A	VA	VIA	VII A	*	He
①	Li	①	Be											*	C	*	*	*	*	Ne
①	Na	①	Mg	IIIB	IVB	VB	VIB	VII B	VIII			IB	II B	Al	Si	*	*	*	*	Ar
①	K	Ca	①	Sc	②	①	①	①	①	①	①	①	①	①	①	①	*	*	*	*
*	Rb	Sr	Y	Zr	①	①	☼	①	①	①	*	①	①	①	①	①	①	①	*	*
*	Cs	③	†	Hf	①	①	①	①	①	*	*	*	*	①	①	☼	☼	☼	☼	
☼	Fr	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	

* = Not a solid at 1000 K

☼ = Radioactive

① = $\text{Si} + \text{MO}_x \rightarrow \text{M} + \text{SiO}_2$

② = $\text{Si} + \text{MO}_x \rightarrow \text{MSi}_y + \text{SiO}_2$

③ = $\text{Si} + \text{MO}_x \rightarrow \text{M} + \text{MSi}_x\text{O}_y$

†	La	Ce	Pr	Nd	☼	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
‡	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼	☼
	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

Requirements

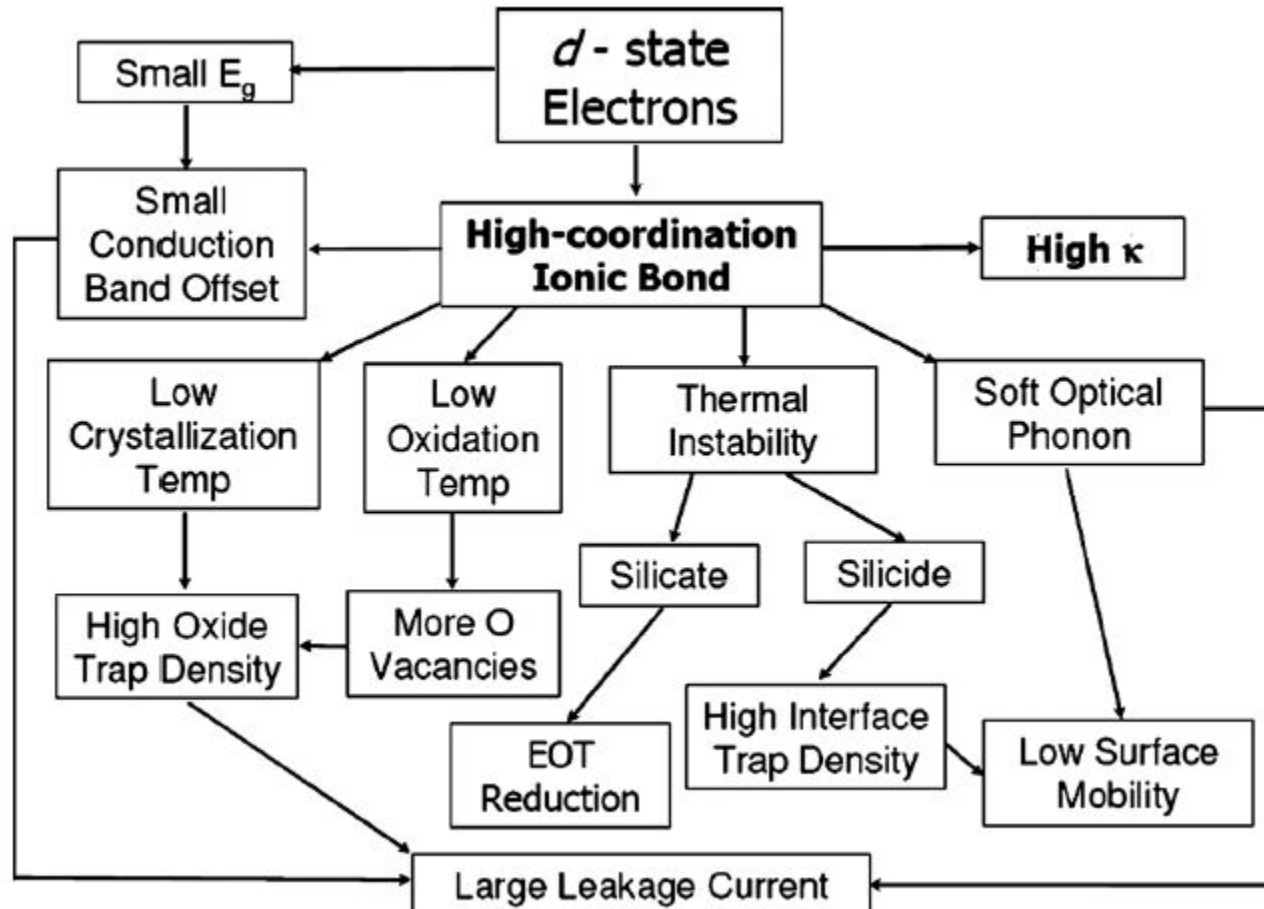
1. It must have a high enough K that it will be used for a reasonable number of years of scaling.
2. The oxide is in direct contact with the Si channel, so it must be thermodynamically stable with it.
3. It must be kinetically stable, and be compatible with processing to 1000 °C for 5 seconds.
4. It must act as an insulator, by having band offsets with Si of over 1 eV to minimise carrier injection into its bands.
5. It must form a good electrical interface with Si.
6. It must have few bulk electrically active defects.

Comparison of diff high k gate dielectrics

Comparison of various characteristics and main features of existing and potentially high- κ gate dielectrics

Dielectric	Dielectric constant (bulk)	Bandgap (eV)	Conduction band offset (eV)	Merits	Drawbacks
Silicon dioxide (SiO_2)	3.9	8.9	3.15	Excellent Si interface, low Q_{ox} and D_{it}	Low- κ , EOT > 0.8 nm
Silicon nitride (Si_3N_4)	7–7.8	5.3	2.1	Good interface and bulk properties, medium Q_{ox} and D_{it}	Low- κ , EOT > 0.5 nm
Aluminum oxide (Al_2O_3)	9–10	8.8		E_g comparable to SiO_2 , amorphous Good thermal stability	Medium Q_{ox} and D_{it} , medium κ
Tantalum pentoxide (Ta_2O_5)	25	4.4	0.36	High- κ	Unacceptable ΔE_C , not stable on Si,
Lanthana (La_2O_3)	~27	5.8	2.3	High- κ , better thermal stability	Moisture absorption, instable with Si
Gadolinium oxide (Gd_2O_3)	~12	~5	— ^a	Low D_{it}	High Q_{ox}
Yttrium oxide (Y_2O_3)	~15	6	2.3	Large E_g	Crystallization
Hafnia (HfO_2)	~20	5.6–5.7	1.3–1.5	Most suitable compared to other candidates	Low crystallization temperature, high D_{it} , silicide formation
Zirconia (ZrO_2)	~23	4.7–5.7	0.8–1.4	Similar to hafnia	Crystallization, silicate and silicide formation, High Q_{ox} and D_{it}
Strontium titanate (SrTiO_3)	~300	3.3	–0.1	High- κ	Marginal stable with Si, crystallization, silicide formation Unacceptable E_g and ΔE_C , field fringing effect

Source of all problems



Bandgap

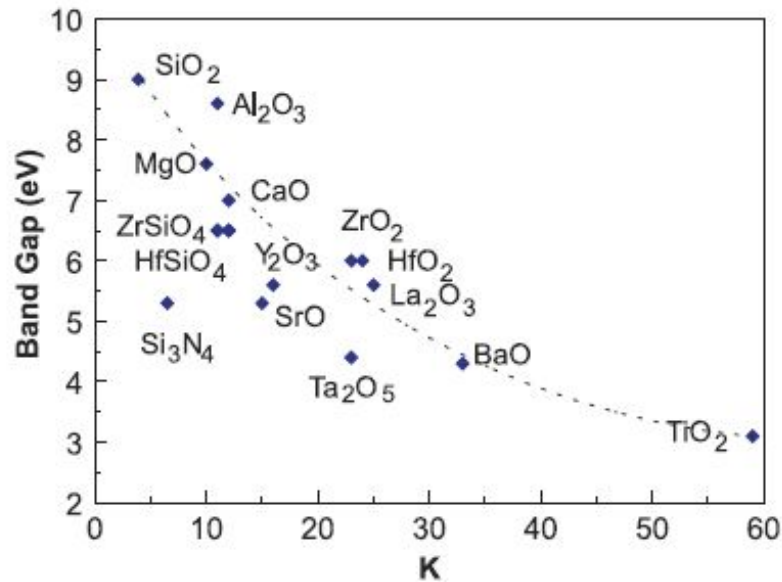


Fig. 5. Static dielectric constant vs. band gap for candidate gate oxides, after Robertson [8].

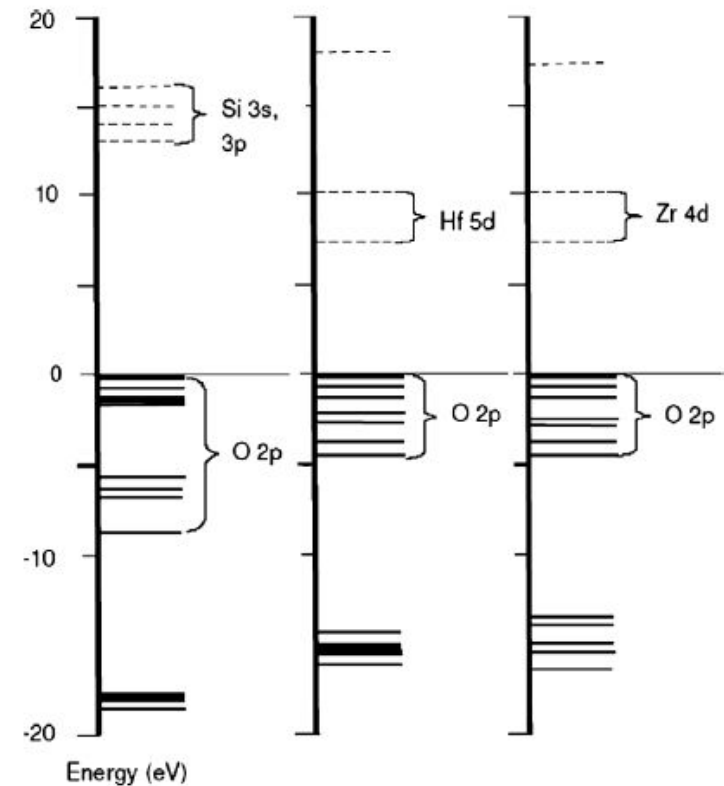
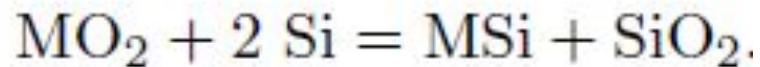
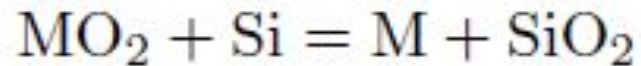


Fig. 39. Calculated energy levels of SiO₂, HfO₂ and ZrO₂. The solid and

Thermodynamic stability



Possible candidates: SrO, CaO, BaO, Al_2O_3 , ZrO_2 , HfO_2 , Y_2O_3 , La_2O_3 and the lanthanides

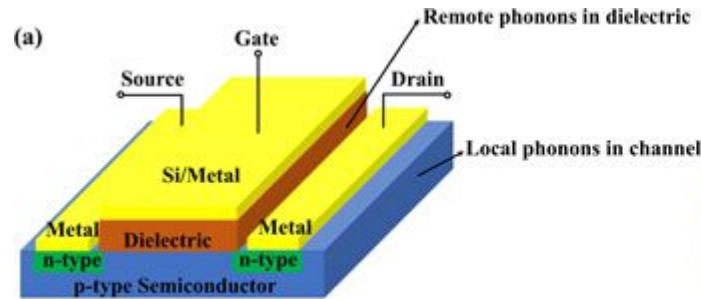
ZrO_2 is more prone to form silicide compared to HfO_2

Comparison of diff high k gate dielectrics

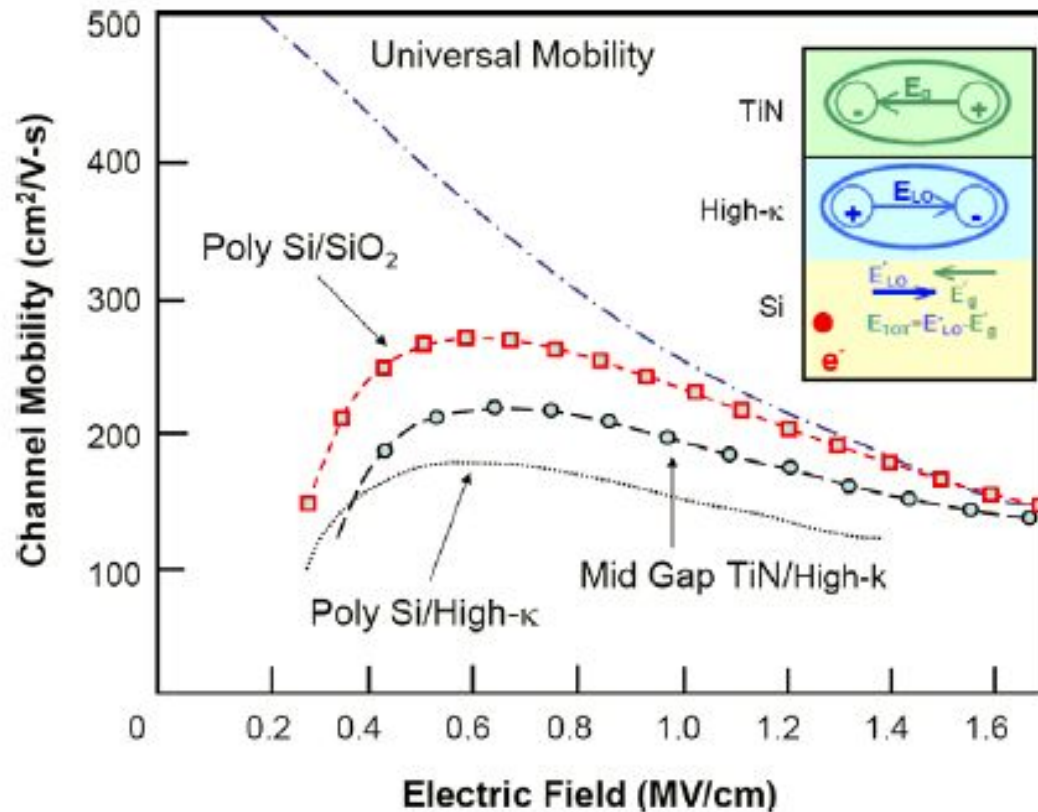
Comparison of the material properties of hafnium oxide and zirconium oxide with silicon oxide

Properties	SiO ₂	HfO ₂	ZrO ₂
<i>Fundamental</i>			
Structure	Amorphous	a, nano-c Crystal ^a	a, nano-c Crystal ^a
Dielectric constant	3.9	20–25	22–25
Bandgap (eV)	8.9	5.6	4.7–5.7
Conduction band structure (Major)	Si 2p	Hf 5d	Zr 4d
Bond length with O (Å)	1.62	1.96 ^b	1.94 ^b
Lattice parameter (Å)	NA	5.116	$a = 5.07, b = 5.16$
<i>Technological</i>			
Formation temperature (°C)	>700	~350	~350
Crystallization temperature (°C)	NA	>350	>350
Stability with Si (kJ/mol) @298 K	Stable	+92.1 ^c	–27.9 ^c
Silicide formation	NA	Yes	Yes
<i>Electrical</i>			
Hole barrier ^d (eV)	4.2	2.5–3.3	2.5–2.9
Electron barrier ^d (eV)	3.15	1.3–1.5	0.8–1.4
Interface trap density (eV ⁻¹ cm ⁻²)	~10 ¹⁰	~10 ¹²	~10 ¹²
Oxide trap density (cm ⁻²)	~10 ¹¹	~10 ¹²	~10 ¹²
Breakdown field (MV/cm)	~10	<4	<4

Mobility

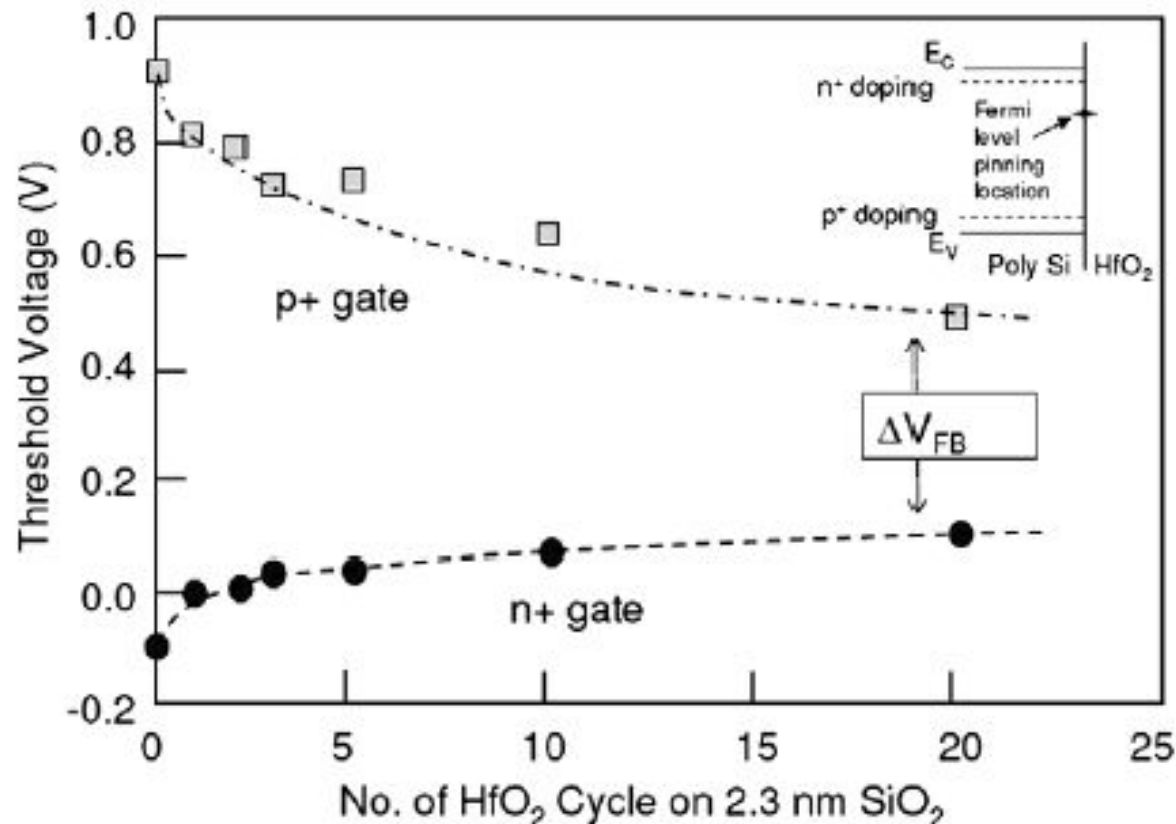


$$\frac{1}{\mu} = \frac{1}{\mu_C} + \frac{1}{\mu_{PH}} + \frac{1}{\mu_{SR}}$$



Additional remote phonon scattering in high k dielectrics. Electrical coupling of metal gate plasmons to dielectric phonons can reduce the phonon scattering with inversion carriers

Fermi level pinning



Hf-Si bonds at the gate electrode interface lead to pinning of the Fermi level of the gate electrode

Most of the metal work functions are in mid gap or close to mid-gap. Needs to be stable under S/D anneal (1050C)

Charge neutrality level (CNL) is an energy level within the energy gap of the semiconductor, which marks the transition of donor like surface states to acceptor like surface states in the energy ladder. Depending on the semiconductor surface preparation and chemical structure, the equilibrium Fermi energy can coincide with the CNL if the density of surface states or interface traps is very large around CNL, the states can trap charge carriers being transferred to/from bulk semiconductor to metal (due to Fermi level difference) interfacing with it. The coincidence of CNL with equilibrium Fermi level is also known as Fermi level pinning phenomenon.

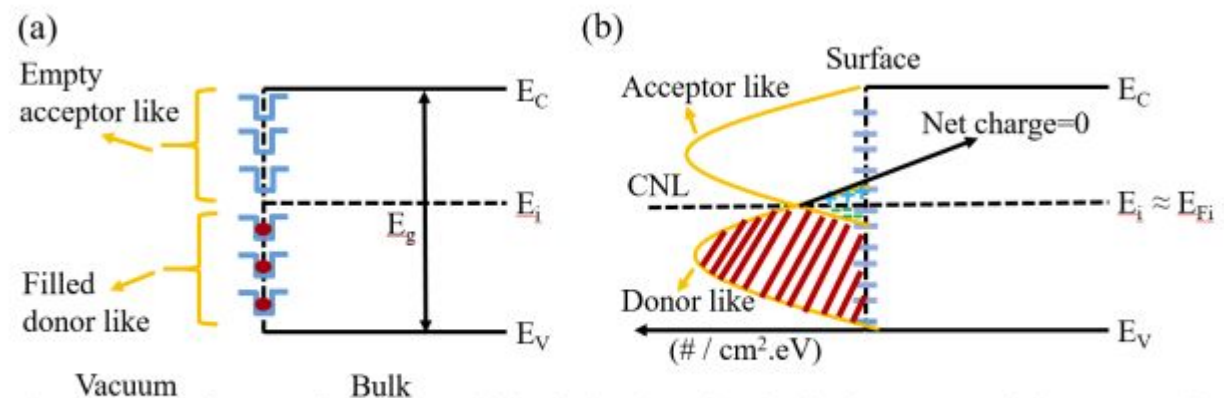
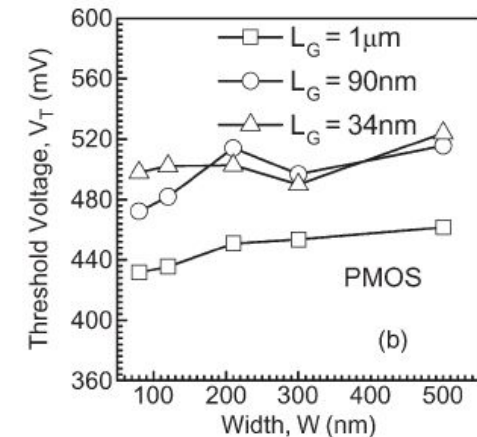
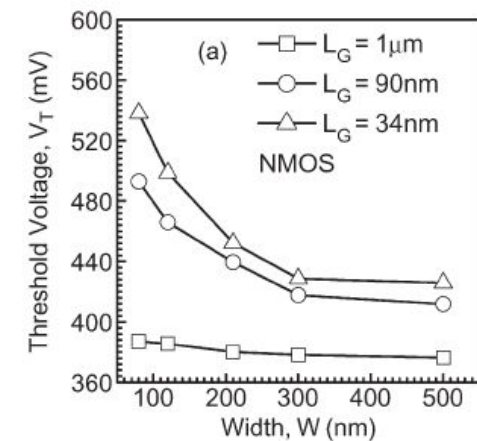


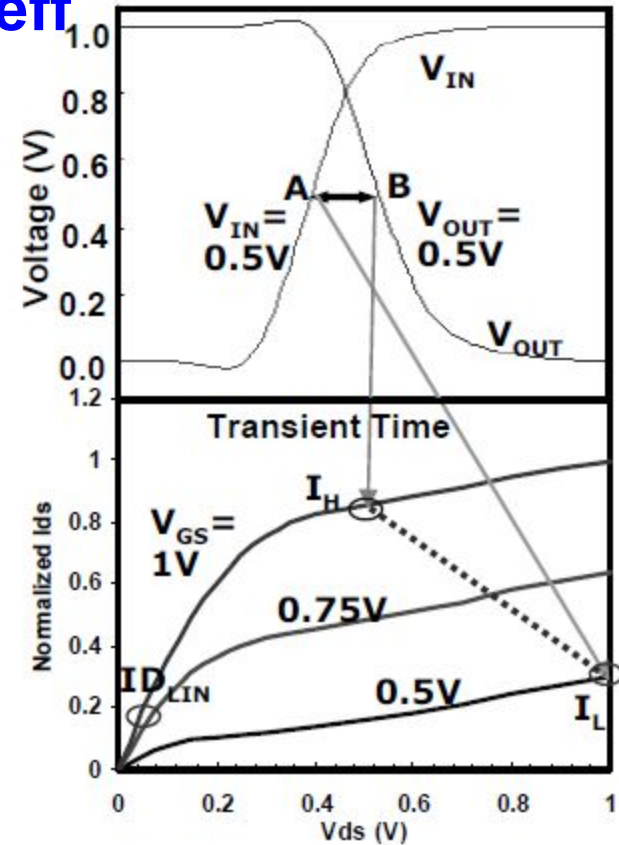
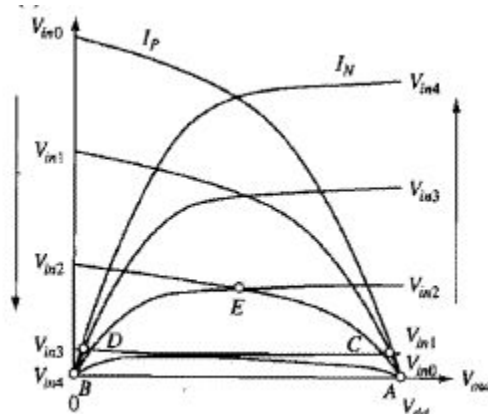
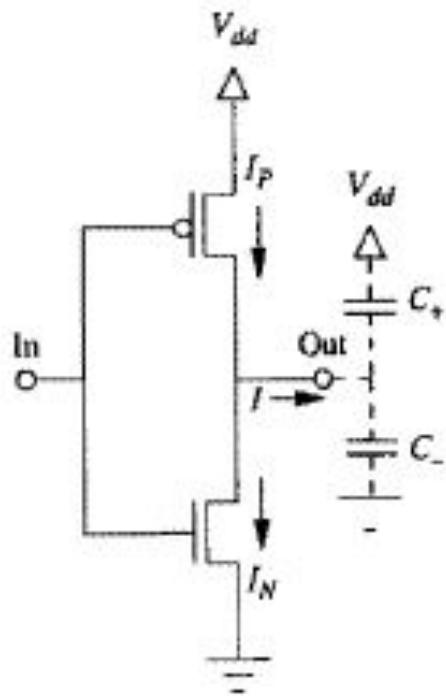
Fig. 2: (a) Interface trap distribution with in the bandgap showing both acceptor and donor types, (b) Representation of D_{it} with Gaussian distribution along with a charge neutral level (E_{CNL}) [2].

Intrinsic oxygen vacancies

- As deposited oxide films are found to have large amount of oxygen vacancies
- Trap electrons and causes V_T instability
- Anomalous width effect (V_T rolls up for narrow devices)



CMOS Inverter: What is I_{eff} ?



I_{eff} is the average switching current = $(I_H + I_L)/2$

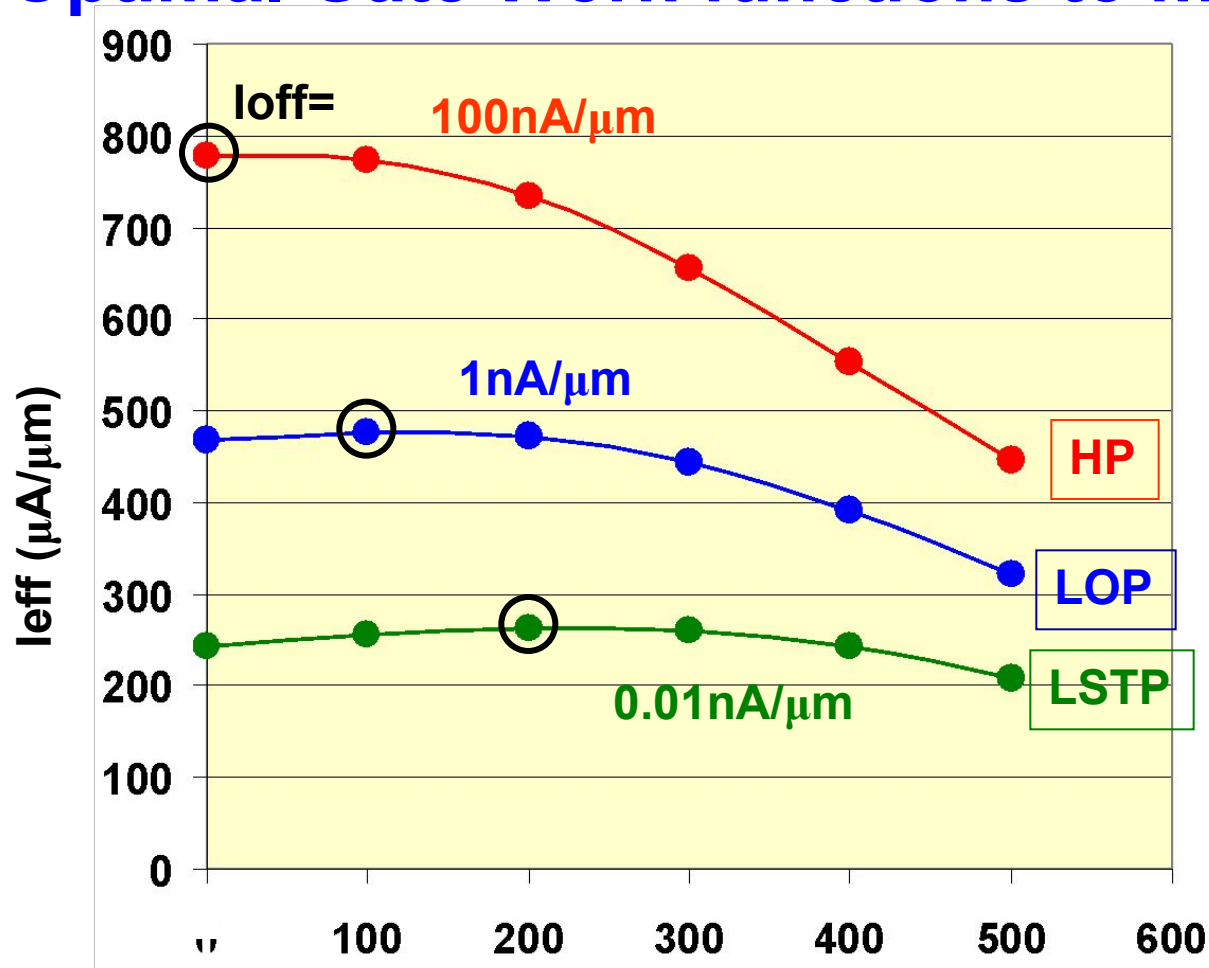
Ring AC Reff: $C_{eff} = 2 * \text{Delay} * (I_{DDA} - I_{DDQ}) / V_{DD}$

AC $R_{EFF} = 2 * \text{Delay} / C_{eff}$

DC Reff based on I_{EFF} : $I_{EFF} = (I_H + I_L) / 2$

DC Reff = $(V_{DD} / N I_{EFF} + V_{DD} / P I_{EFF}) / 2$

Optimal Gate Work-functions to maximize I_{eff}



- Optimal gate work-function depends on the technology V_t/I_{off} targets.
- If $WF < \text{optimal}$: high doping degrades mobility
- If $WF > \text{optimal}$: lower doping results in poor gate control and degraded SS.

Si conduction
band-edge

Increasing work-function (meV)
→

Gate 1st vs Gate Last

Gate-First

Dep Hi-k &
Met 1



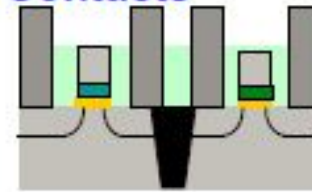
Patt Met 1 &
Dep Met 2



Patt Met 2 &
Etch Gates



S/D formation &
Contacts

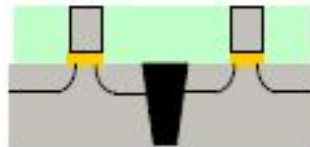


Hik-First, Gate-Last

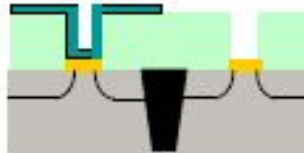
Dep & Patt
Hik+Gate



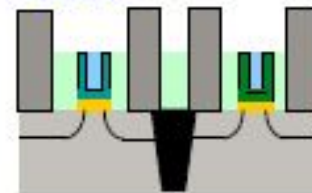
S/D formation &
ILD dep/polish



Rem Gate &
Patt Met 1



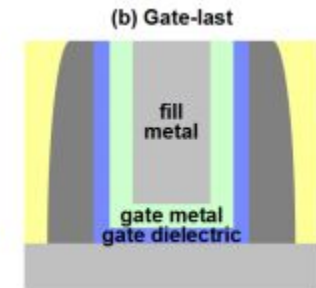
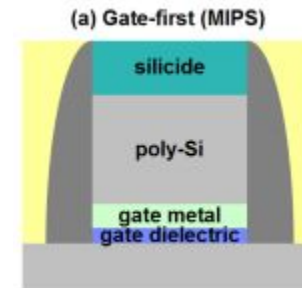
Dep Met 2+Fill &
Polish



Gate 1st vs Gate Last: process flow



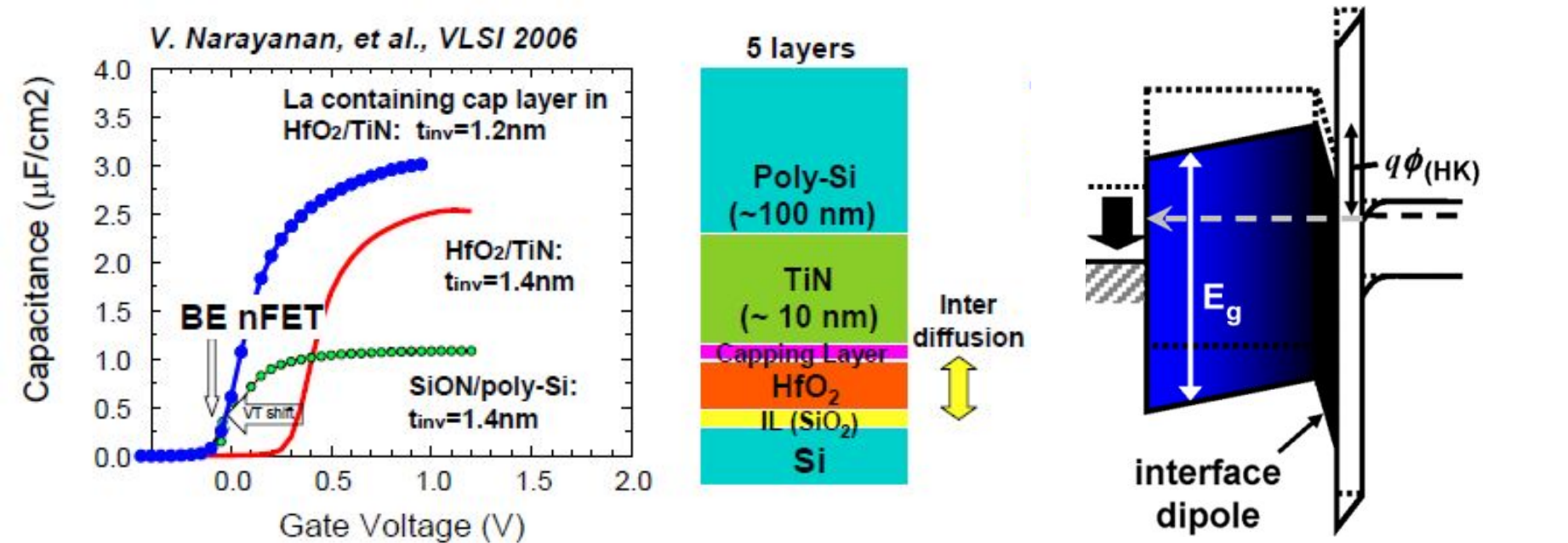
- Isolation
- Gate stack dep IL-SiO₂/HfO₂ (+ ESL)/a-Si [HKF]
- Gate patterning Dummy gate = SiON or SiO₂/a-Si [HKL]
- Halos/Extensions
- Spacers + HDDs + RTA
- S/D silicidation (NiPtSi)
- CESL + ILD0 dep/CMP **RMG module**
- Dummy poly-Si gate removal
- Dummy dielectric removal + IL-SiO₂/HfO₂ dep [HKL]
- WF-MG + fill-MG(W) dep/CMP
- ILD1 dep + W-filled contacts formation
- BEOL



Tradeoffs

Gate First	Gate Last
Structurally more closer to poly/SiON	Quite different
Gate stack deposition before S/D anneal	High K first Metal Gate last Both High K and Metal Gate Last
Less choice of metal electrodes	More choice (Realizing PFET work function is easier)
No restriction	Restriction on layouts (poly open, CMP)
	Metal gate stress from dummy poly open
Poly on top of metal	No poly on top of metal

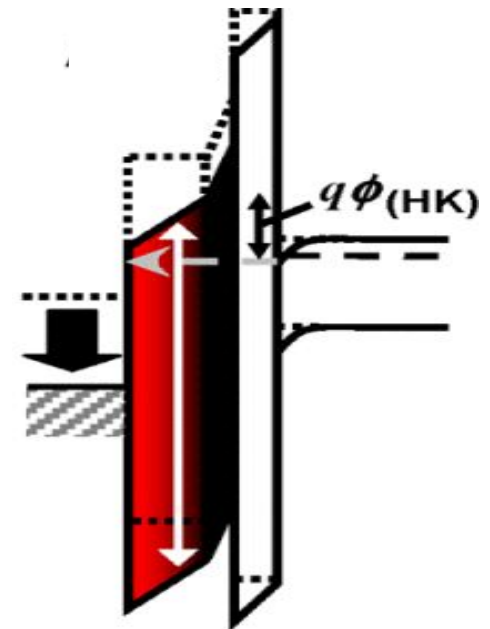
Gate 1st mechanism (IBM/GF/Samsung 32/28nm) 30

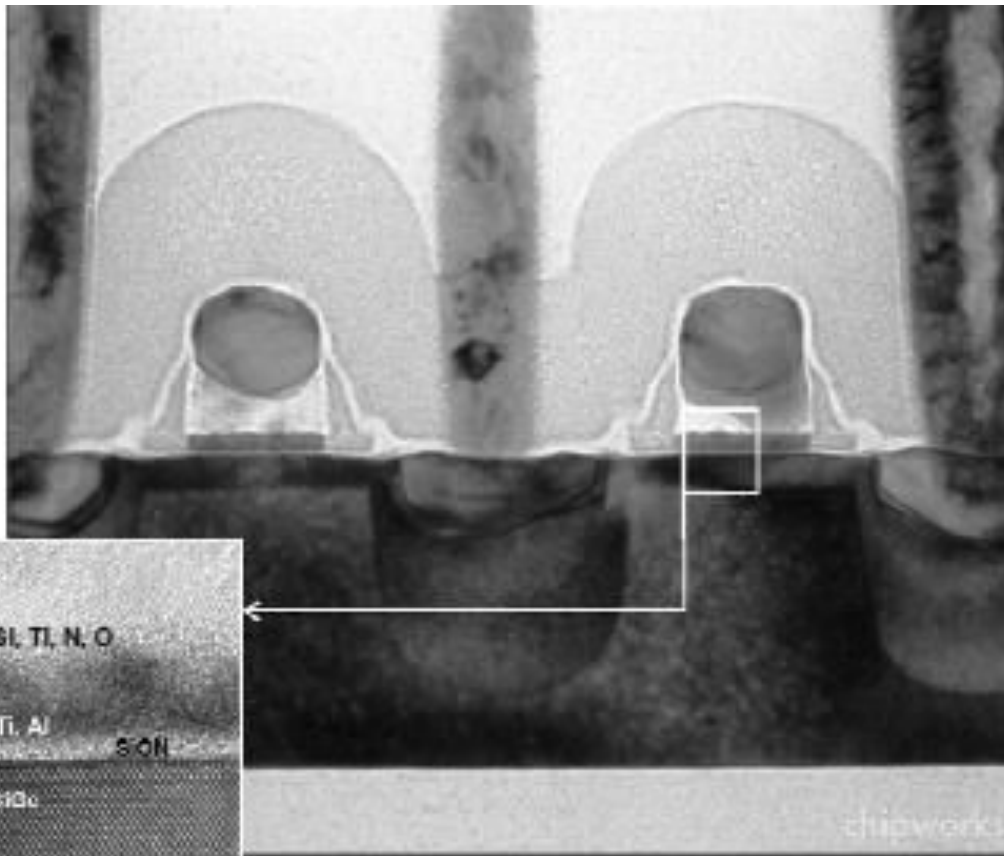


$$\Delta V_{\text{dipole}} = \frac{Q_{\text{dipole}} \times N_{\text{dipoles}} \times d}{\epsilon_{\text{dipole}}}$$

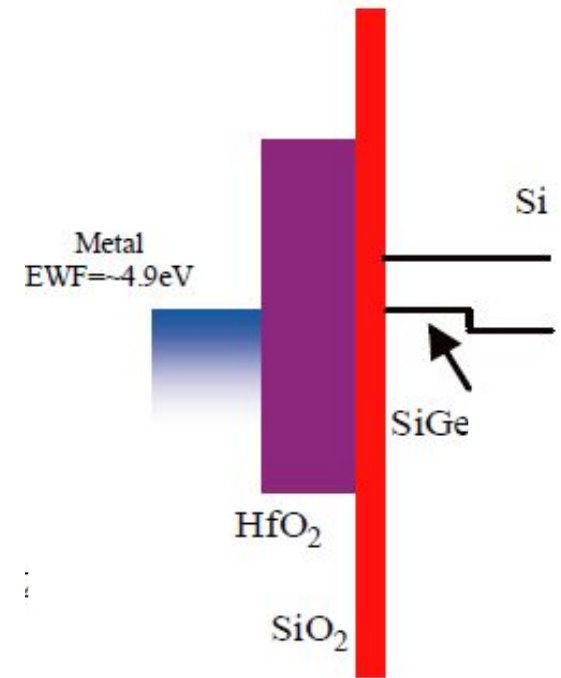
$$\Phi_{\text{m,eff}} = \Phi_{\text{m,vacc}} + \Delta V_{\text{dipole}}$$

$$V_{\text{FB}} = \Phi_{\text{m,eff}} - Q_{\text{f}} \frac{\text{EOT}}{\epsilon_{\text{ox}}} - \Phi_{\text{s}}$$





Apple A7 processor in iPhone 5S
(Samsung 28LP)



Single metal dual dielectric dual channel

La dipole infused HfO_2 – NFET

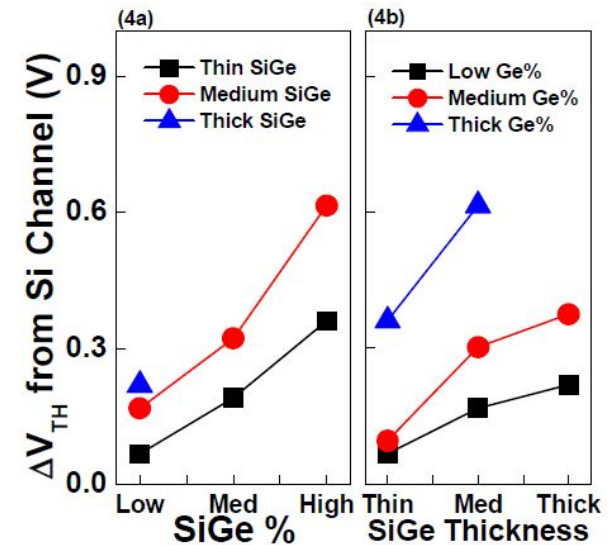
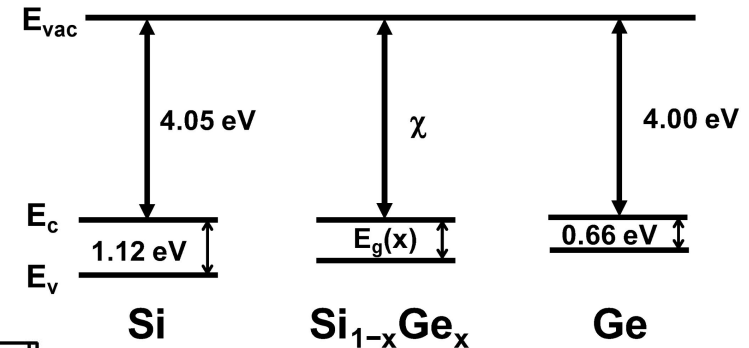
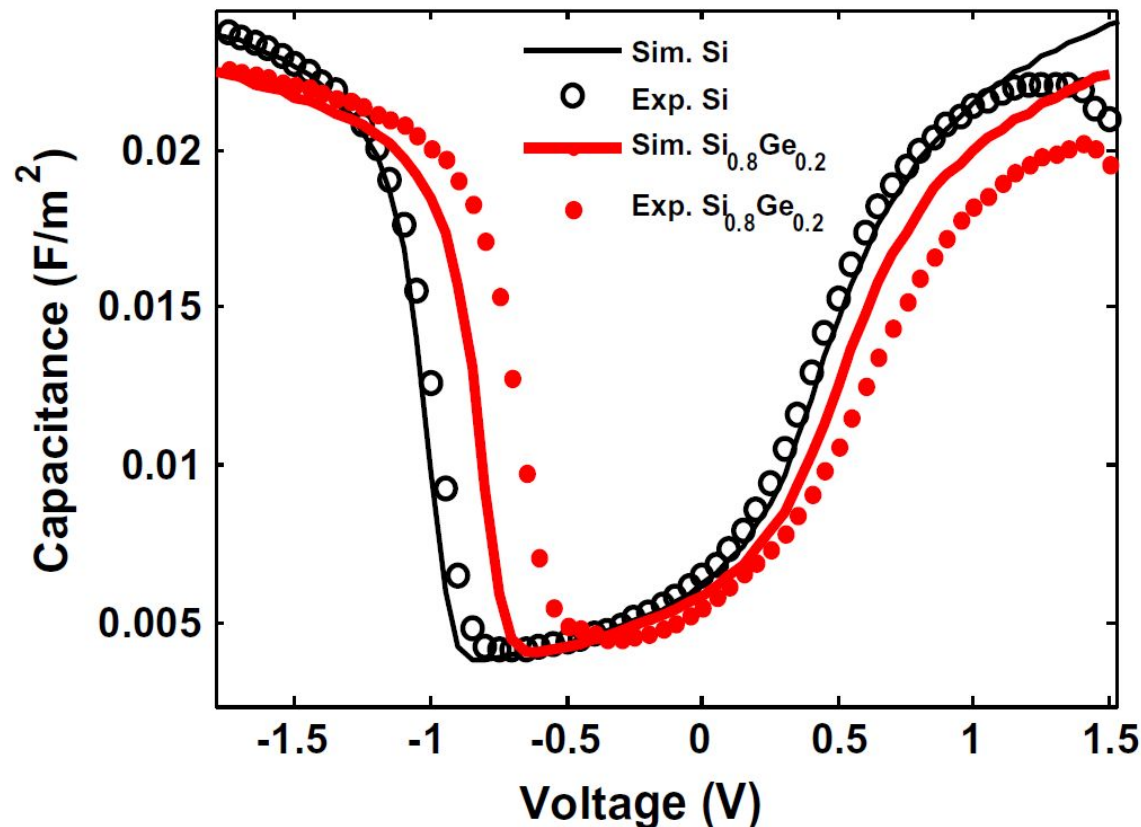
HfO_2 and SiGe channel (biaxially compressed) – PFET*

Lin JAP '11

*Krishnan IEDM '11

SiGe channel enabled PFET w/ low V_T

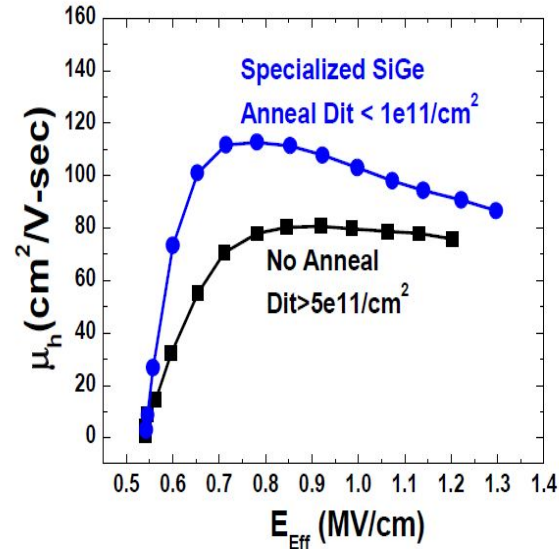
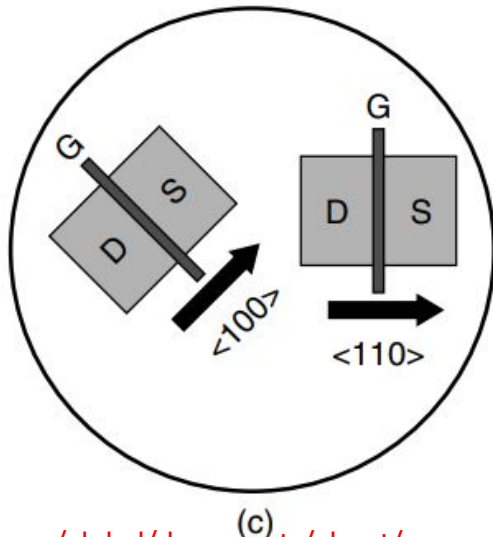
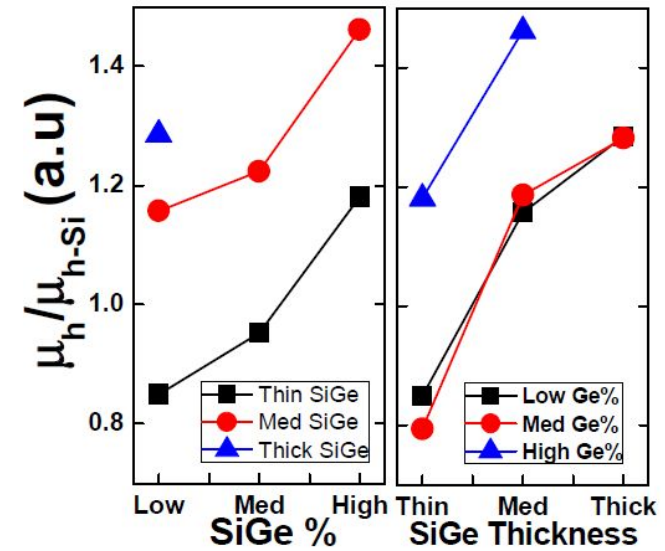
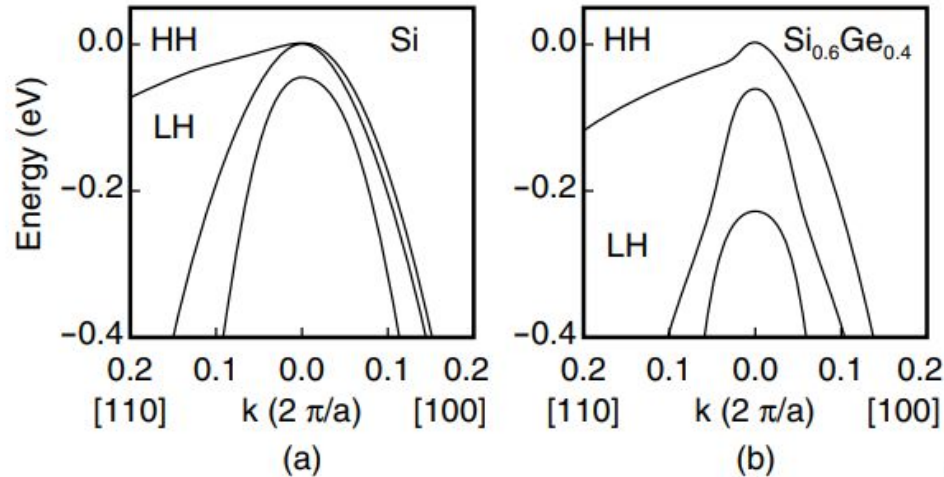
Absence of capping layer which can cause large V_T shift in PFETs by dipole formation



[A. Soussou et al. / Microelectronic Engineering 109 \(2013\) 282–285](#)

[S. Krishnan et al., "A manufacturable dual channel \(Si and SiGe\) high-k metal gate CMOS technology with multiple oxides for high performance and low power applications," 2011 International Electron Devices Meeting, Washington, DC, USA, 2011](#)

Higher hole mobility in SiGe channel (biaxial compression)



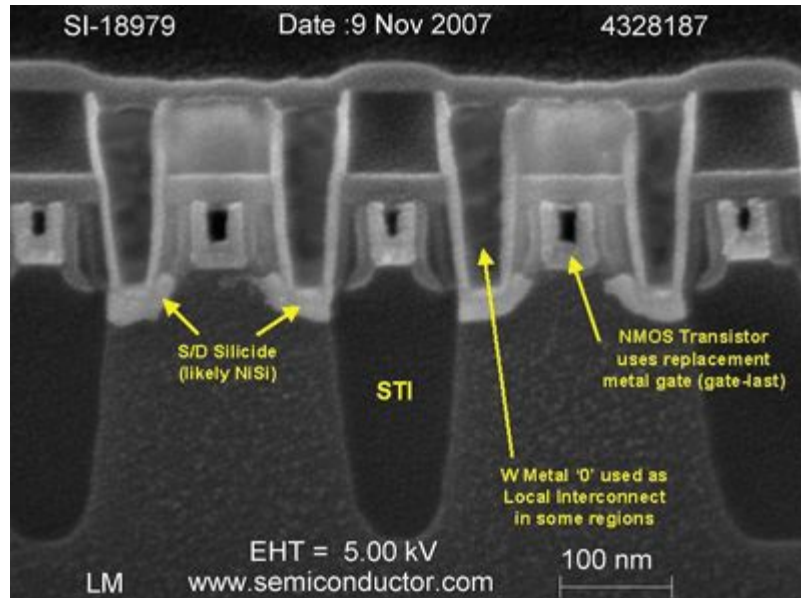
High-pressure
Deuterium anneal (1h,
400 deg. C, 20 atm)*

<https://www.fujitsu.com/global/documents/about/resources/publications/fstj/archives/vol39-1/paper10.pdf>

S. Krishnan *et al.*, "A manufacturable dual channel (Si and SiGe) high-k metal gate CMOS technology with multiple oxides for high performance and low power applications," 2011 International Electron Devices Meeting, Washington, DC, USA, 2011

*<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=7223654>

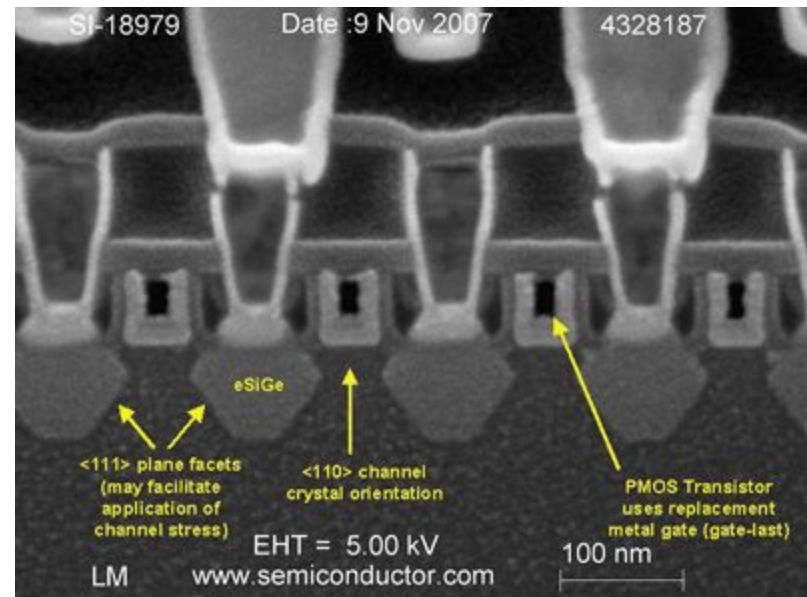
Intel Penryn (High k first Metal Gate last - 45nm)



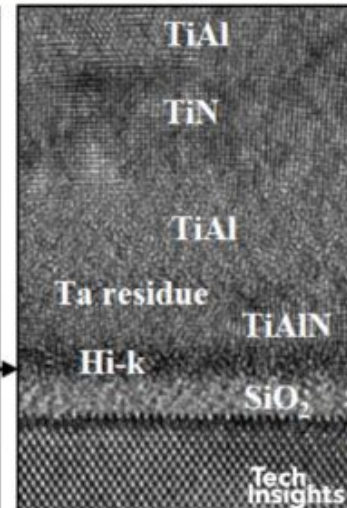
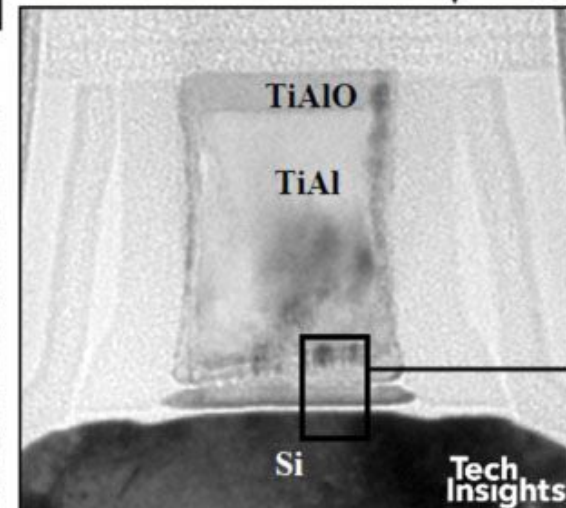
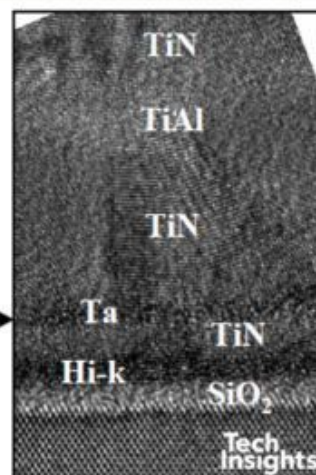
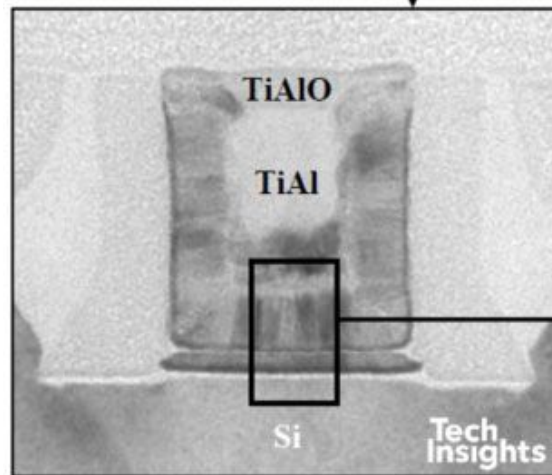
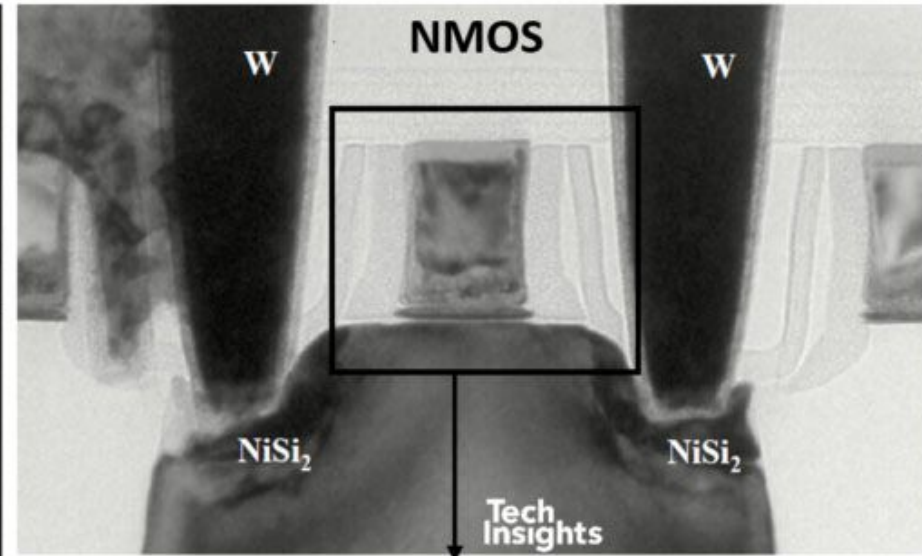
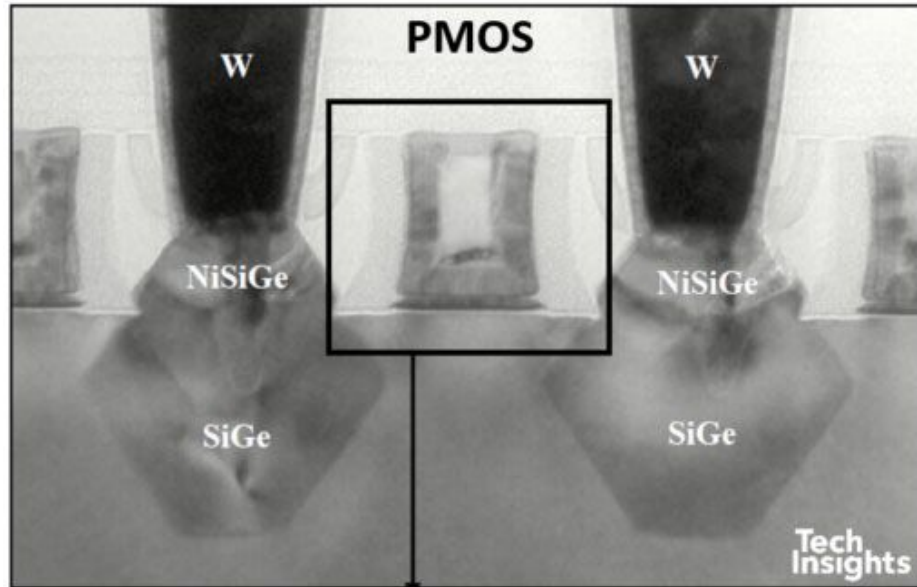
Single dielectric dual metal

TiN – PFET metal

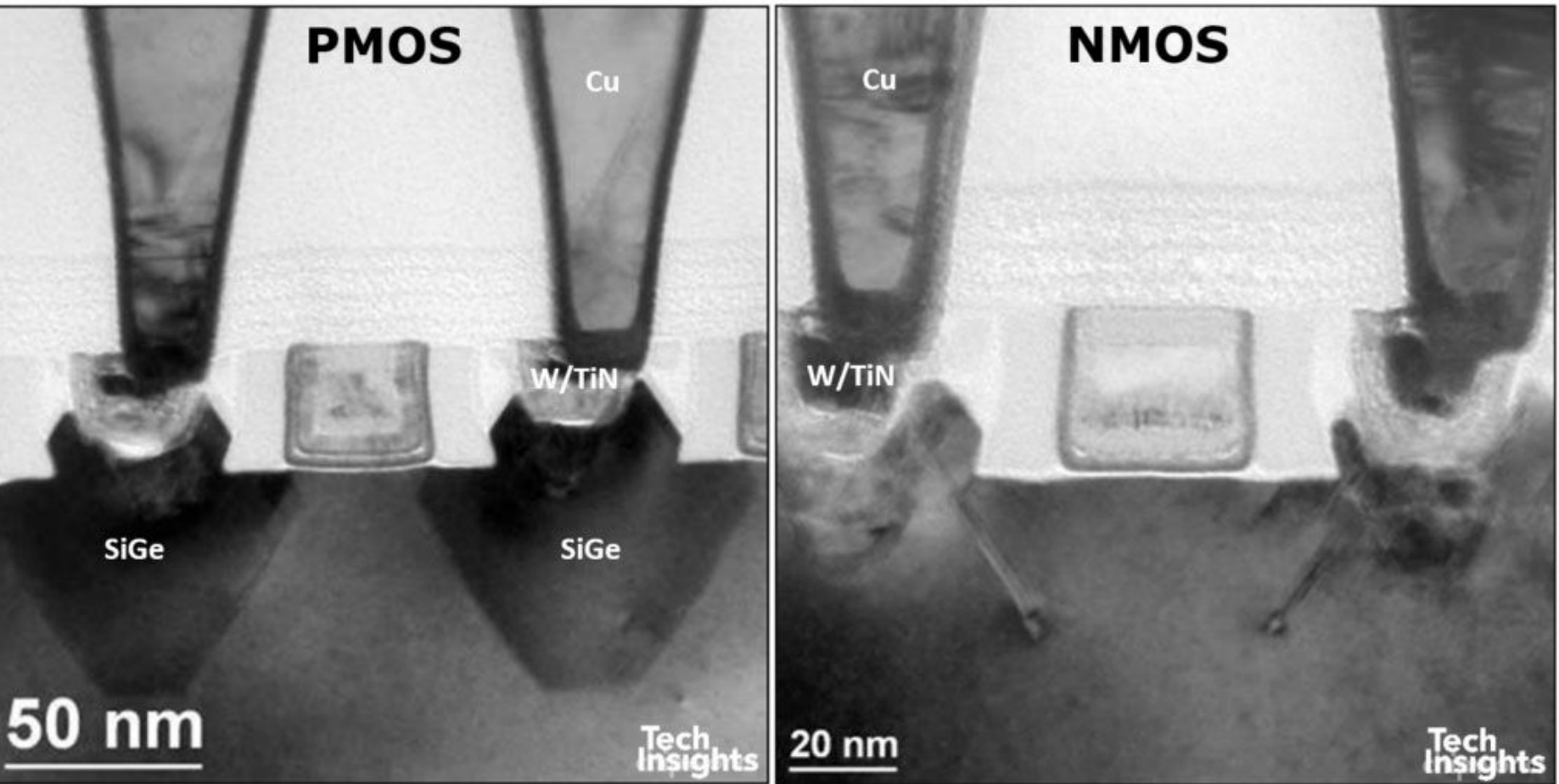
TiAlN – NFET metal



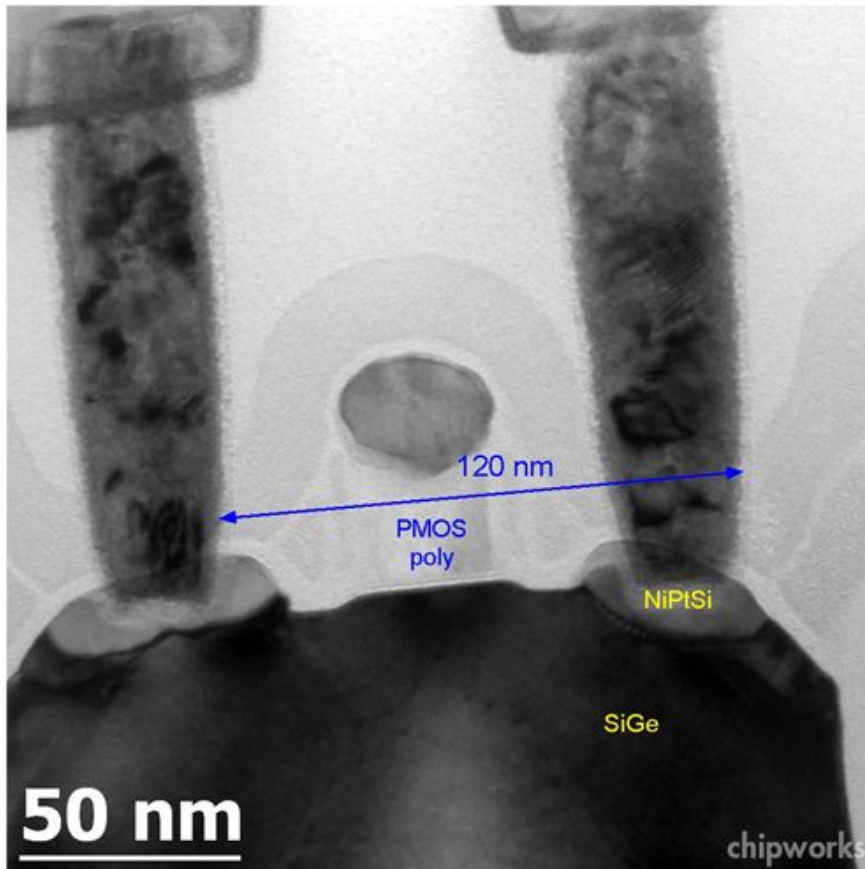
Intel Penryn (High k first Metal Gate last - 45nm)



Intel Penryn (High k last Metal Gate last - 32nm)



TSMC 28nm process



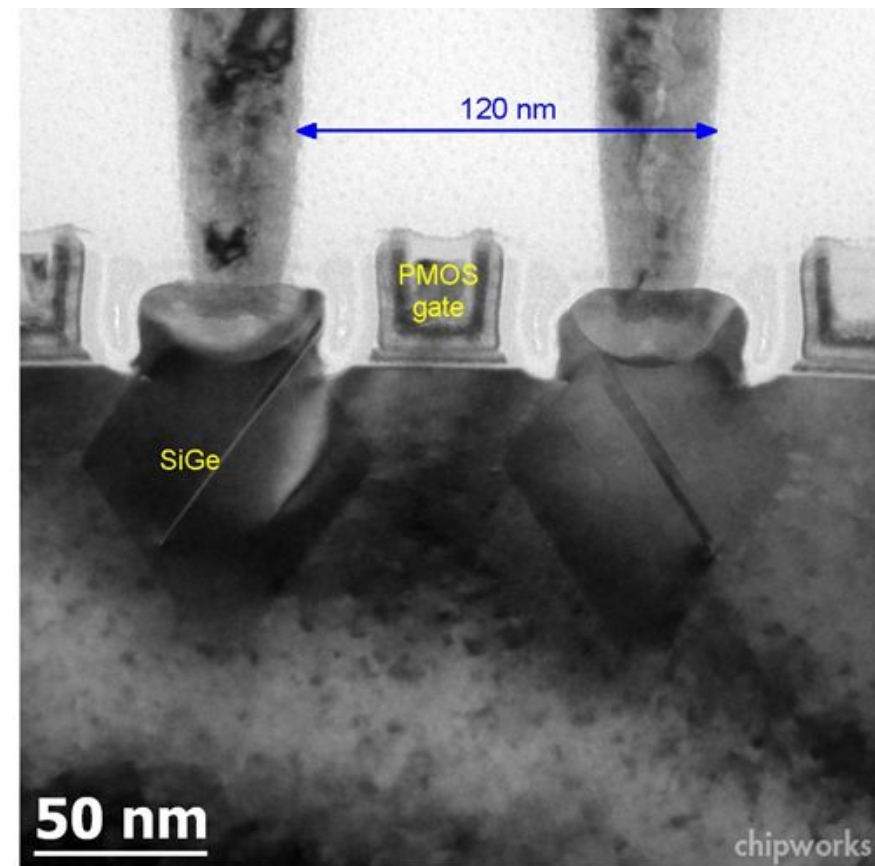
TSMC had two versions of gate stack for 28nm node - poly/SiON (28LP for Qualcomm) and HKMG

Standard Offerings	HP	HPM	HPL	LP
Core Vcc (V)	0.85	0.9	1.0	1.05

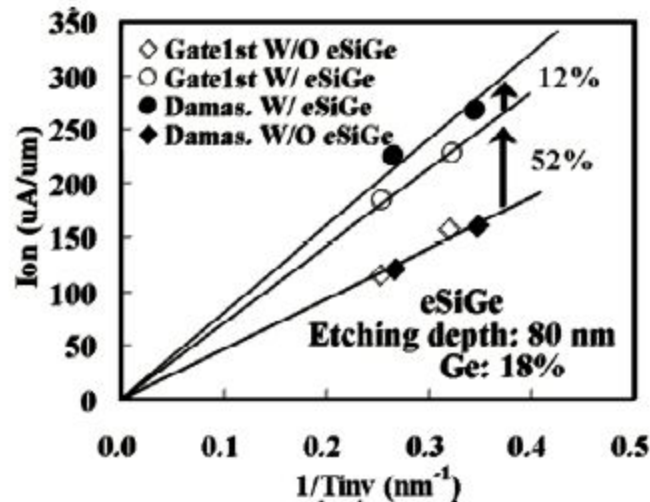
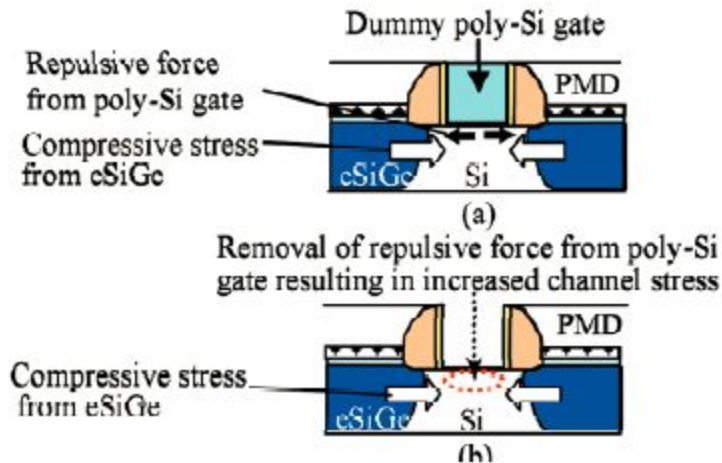
HPL- HKMG w/o eSiGe

HP - HKMG w/ eSiGe

LP - Poly/SiON w/ eSiGe



Enhanced PMOS strain by RMG



Wang – Sony
VLSI 2007 [86]

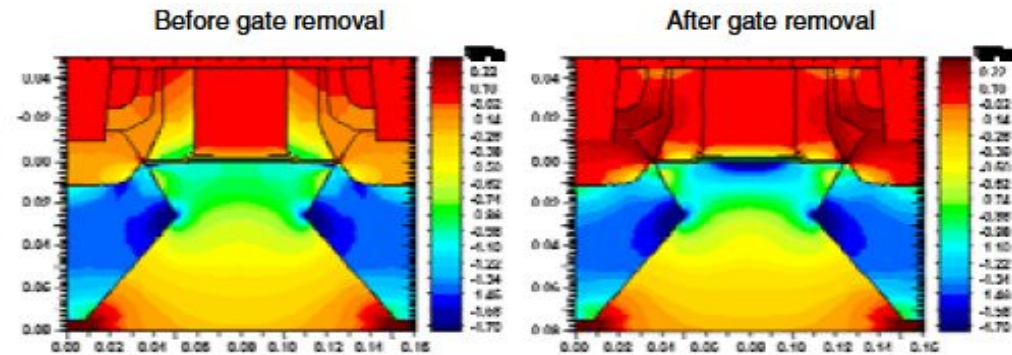
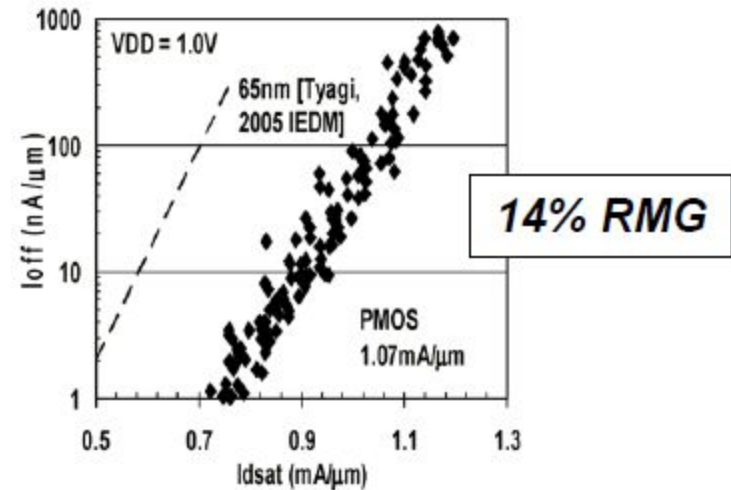
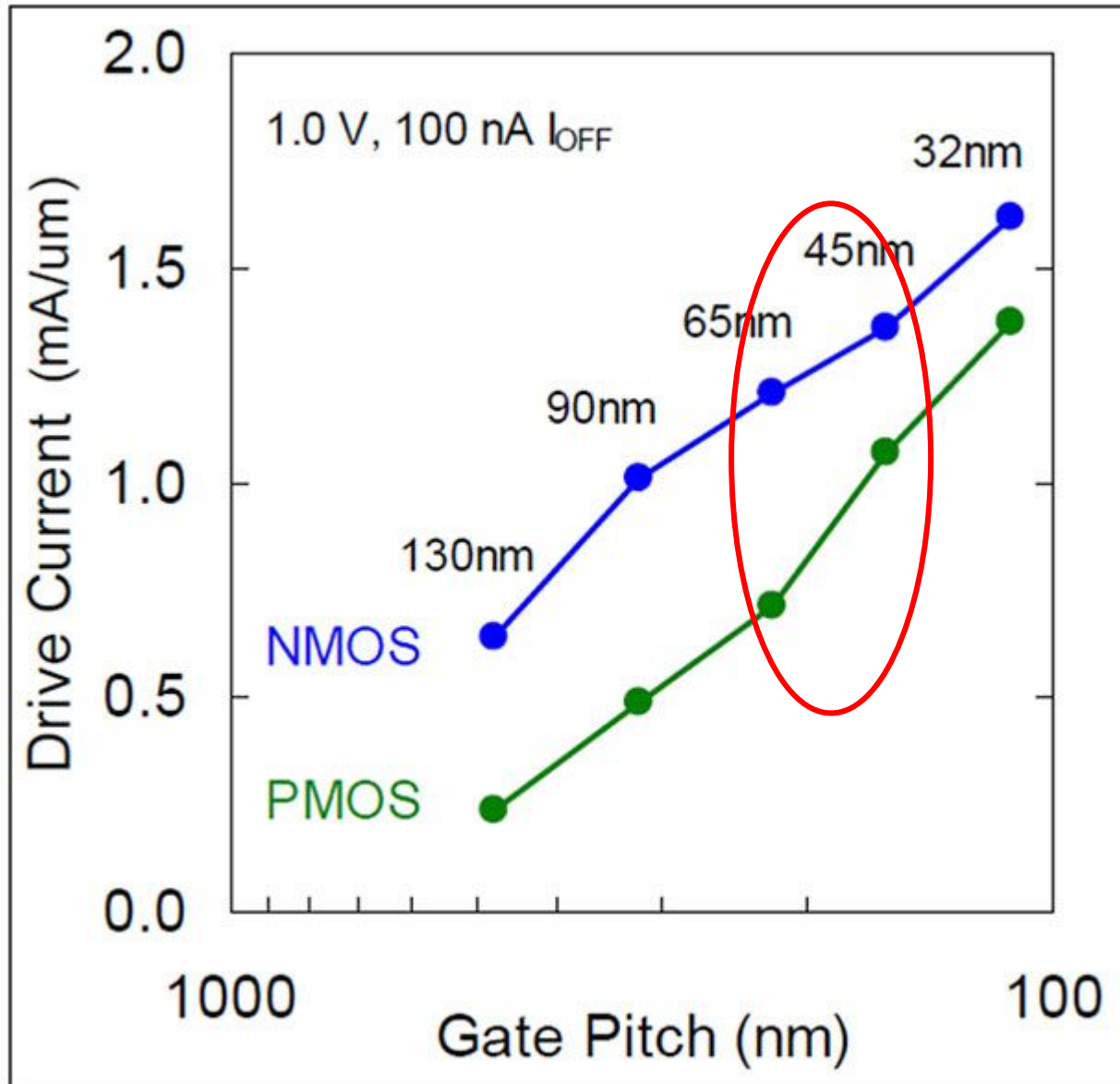


Figure 17: Stress contours in the PMOS transistor before and after the removal of the polysilicon dummy gate. Stress in the channel is shown to increase 50% from 0.8GPa to >1.2 GPa.



Auth – Intel
VLSI 2008 [8]



PMOS drive current increases noticeably from the 65nm to the 45nm node

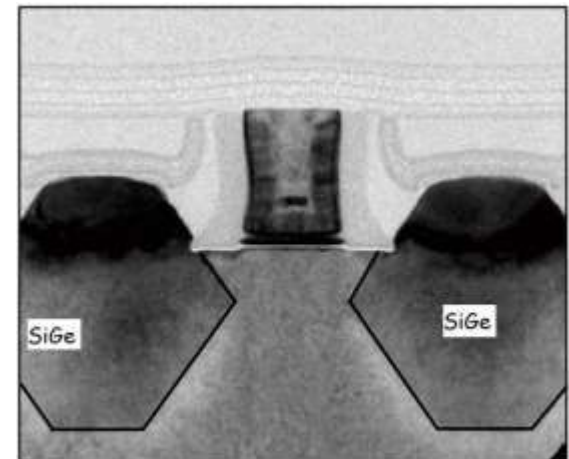
Intel IEDM 2009

Try these two problems

The p^+ silicon substrate of an MOS capacitor is connected to ground. The oxide thickness is 5 nm. The workfunction of the gate electrode metal is same as that of an n^+ polysilicon. Assume the equilibrium Fermi level in isolated n^+ and p^+ Silicon coincide with E_C and E_V respectively. Draw the energy band diagram (showing E_C , E_i , E_V and E_F) and electrostatic potential along the vertical cut (Gate-oxide-semiconductor) for these three different cases

- Flat-band** in p^+ silicon with no charge in oxide or at any interface
- Flat-band** in p^+ silicon with a 2-D charge of density $-1.1 \mu\text{C}/\text{cm}^2$, is present 1.5 nm away from the gate-oxide interface for the same structure.
- Flat-band** in p^+ silicon with a 2-D electric dipole with a dipole moment, $0.385 \text{ pC}/\text{cm}$ (directed towards gate/oxide interface) is present 1.5 nm away from gate-oxide interface for the same structure.

Shown on the right is TEM cross section of a PFET from Intel's 45nm process. The source/drain region is etched and filled with SiGe to boost hole mobility. If you are given a silicon wafer till gate stack is patterned, give process steps by which you can create such a trench for SiGe fill. The wafer orientation is $\langle 001 \rangle$ and channel orientation is $\langle 110 \rangle$



References

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3. J. Robertson, “High dielectric constant oxides”, *Eur. Phys. J. Appl. Phys.* 28, 265–291 (2004)
4. J. Robertson, ”High dielectric constant gate oxides for metal oxide Si transistors”, 2006 *Rep. Prog. Phys.* 69 327
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6. Yuan Xiao Ma, Hui Su, Wing Man Tang, Pui To Lai; Review on remote phonon scattering in transistors with metal-oxide-semiconductor structures adopting high-k gate dielectrics. *J. Vac. Sci. Technol. B* 1 November 2023
7. M. H. Na, E. J. Nowak, W. Haensch and J. Cai, "The effective drive current in CMOS inverters," *Digest. International Electron Devices Meeting,,* San Francisco, CA, USA, 2002, pp. 121-124